

A PROJECT REPORT ON

**Bridging Language Barriers: The Role of AI in
Real-Time Multilingual Translation for Video
Conferencing**

**SUBMITTED TO THE SAVITRIBAI PHULE UNIVERSITY, PUNE
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FOR THE AWARD OF THE DEGREE**

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2024-2025**



CERTIFICATE

This is to certify that the project report entitled

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*It gives us great pleasure in presenting the project report on ‘**Bridging Language Barriers: The Role of AI in Real-Time Multilingual Translation for Video Conferencing**’.*

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ABSTRACT

This paper reports on a real-time multilingual speech-to-speech translation and transcription system deployed as part of a video conferencing application. Deepgram speech-to-text is used, and translation is handled using the Google Translate API. Pusher maintains smooth real-time updates, with LiveKit dealing with high-fidelity audio and video streaming in real time with little latency. The solution presented greatly mitigates language barriers, allowing smooth collaboration among speakers of different languages. This project showcases a feasible, scalable solution for enhancing communication in multilingual, diverse settings, opening the door to further innovations in AI-driven conferencing technology. It dynamically processes audio input, converts it into text, translates it into the desired language, and plays back the translated speech in real time. The architecture is highly scalable and optimized for minimal latency, ensuring a responsive and intuitive user experience. This project lays the foundation for future advancements in AI-driven video conferencing platforms, improving inclusivity and collaboration in multilingual scenarios.

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LIST OF ABBREVIATIONS

Abbreviation	Illustration
AI	Artificial Intelligence
API	Application Programming Interface
UI	User Interface
ML	Machine Learning
SDP	Session Description Protocol
SFU	Selective Forwarding Unit
WEBRTC	Web Real-Time Communication

CHAPTER 1

INTRODUCTION

1.1 Overview

With the recent world, this is the age of globalization, and the need for smooth communication is a must, even more so with real-time video conferencing where users from different linguistic backgrounds work together. With this requirement, our project combines the latest technologies to develop a multilingual, real-time audio translation and transcription system integrated in a video conferencing platform constructed with the MERN (MongoDB, Express.js, React, Node.js) stack. The system achieves high accuracy and efficiency by utilizing Deepgram for speech-to-text transcription and Google Translate API for real-time language translation. LiveKit manages low-latency video and audio streaming, maintaining smooth communication with minimal delay between various geographical locations. Pusher manages real-time event updates, making it possible for all participants to receive translated content and transcription at the same time. Supabase is also utilized for safe data storage and user authentication. The platform dynamically records, transcribes, translates, and plays back audio content in several languages, allowing participants to participate more effectively. Adaptive bitrate management and real-time event processing enable the system to operate at high performance even under fluctuating network conditions. The integration of features like real-time transcription, auto-detection of languages, and instant translation increases collaboration, and the system is suitable for remote teams, educational sites, and corporate communication. The scalable architecture of the project ensures that the system can manage multiple participants and various linguistic interactions effectively without compromising on quality. The blending of the technologies gives rise to a system that not only interprets speech but also retains contextual correctness, so that conversations are preserved in their true sense. Capable of transcribing, translating, and providing audio output in real-time, the platform enables users to carry out cross-language conversations with ease. Moreover, the system has been developed to support multiple participants at a time, thus being apt for mass application in virtual conferences, webinars, and international team meetings.

1.2 Motivation

The system supports effortless multilingual real-time communication during video conferencing through automatic spoken language translation into several target languages. It improves the user experience by eliminating the language barrier, thus making global communication more efficient and inclusive. It allows users to communicate in their preferred language. In addition to audio translation, chat translation and caption is also supported for an improved meeting experience. The objective involves combining Deepgram for speech-to-text transcription and Google Translate for translation of the transcribed text, ensuring that the spoken content is converted into text with precision and translated into the preferred language. LiveKit takes care of real-time audio and video streaming, whereas Pusher handles real-time event handling and synchronizing among the participants. This integration of technologies not only makes communication swift and effective but also inclusive so that people with different linguistic backgrounds can work together seamlessly.

1.3 Problem Definition and Objectives

Language barriers in video conferencing hinder effective communication between users from different linguistic backgrounds. Existing platforms lack real-time multilingual support for both audio and chat, leading to delays and miscommunication. There is a need for an AI-powered solution that enables seamless, real-time translation of speech and text, along with essential conferencing features like screen sharing, recording all within a single platform.

1.4 Project Scope & Limitations

The primary scope of this project is to develop a real-time multilingual video conferencing system that enables seamless communication across language barriers through automatic speech recognition (ASR), machine translation (MT), and text-to-speech (TTS) synthesis. The system is designed to support high-quality video calls, real-time audio translation, multilingual chat translation, screen sharing and meeting transcription. Key technologies employed include WebRTC for media transmission, Deepgram for speech-to-text processing and Google Web Speech API for speech synthesis.

The project aims to:

- Facilitate real-time communication between users speaking different languages.
- Provide accurate, low-latency audio and text translation.
- Ensure accessibility through captions and transcripts.
- Offer scalability to support up to 100 concurrent users.

Limitations

Despite the robust feature set, several limitations exist:

- The quality of translation heavily depends on the accuracy of speech recognition and the capabilities of third-party APIs, which may not perform uniformly across all languages or accents.
- Latency issues may arise due to the processing time required for speech recognition, translation, and synthesis, especially under high network load.
- Real-time performance may degrade on low-bandwidth or unstable network connections.

1.5 Methodologies of Problem solving

To address the challenges of real-time multilingual communication in video conferencing, a structured and systematic approach was followed in the design and implementation of the proposed solution. The development methodology integrates multiple technologies and frameworks to build a robust and scalable system capable of real-time speech translation and transcription. The key stages of problem solving are outlined below:

1. Requirement Analysis

The initial step involved identifying the core challenges faced in multilingual communication, particularly in video conferencing platforms. This included the need for:

- Real-time speech recognition.

- Accurate and fast translation.
- Audio output in the user's native language.
- Caption generation and chat translation.

User expectations, technical feasibility, and performance constraints were thoroughly analyzed.

2. System Architecture Design

A modular and scalable architecture was designed based on microservices and real-time communication protocols. Key technologies chosen include:

- **LiveKit** for real-time video and audio streaming.
- **Deepgram API** for automatic speech recognition (ASR).
- **Google Web Speech API** for real-time speech synthesis.
- **Pusher** for real-time event handling and communication.
- **Next.js and TypeScript** for frontend development.
- **Supabase (MySQL)** and Prisma ORM for database operations.

3. Implementation

The system was implemented in the following stages:

1. Capturing user audio using WebRTC.
2. Converting the audio to text using Deepgram's ASR engine.
3. Translating the recognized text into the target language.
4. Synthesizing the translated text back into speech using the Google Translate API.
5. Delivering translated audio to participants while displaying captions and chat in their preferred languages.

4. Real-Time Communication Handling

The real-time aspect of audio and video communication is managed through LiveKit and WebRTC. Socket communication via Pusher ensures that translated messages, captions, and chat are delivered with minimal latency.

5. Testing and Optimization

Extensive testing was conducted to assess:

- Latency between original speech and translated output.
- Translation accuracy across different languages.
- System performance under varying network conditions.

Bugs were identified and fixed, and the system was optimized for responsiveness and scalability.

6. Deployment

The application was deployed using Vercel for the frontend and backend, with a cloud-hosted database via Supabase. The system supports up to 100 concurrent users and delivers seamless multilingual audio and text communication.

7. Evaluation and Feedback

User feedback was collected during testing phases to evaluate usability, accuracy, and overall experience. Continuous improvements were made based on iterative development and feedback loops.

Conclusion

By combining modern web technologies with advanced AI-based translation tools, the proposed methodology ensures efficient, real-time multilingual communication. The solution effectively bridges language barriers in global conversations, enhancing accessibility and inclusivity.

CHAPTER 2

LITERATURE SURVEY

2.1 Literature Survey

Sr. No.	Paper Name	Year	Central Idea
1	RealTranS: End-to-End Simultaneous Speech Translation with Convolutional Weighted-Shrinking Transformer	2021	Uses a hybrid Transformer-convolutional architecture to reduce latency and improve translation in SimulST.
2	How 'Real' is Your Real-Time Simultaneous Speech-to-Text Translation System?	2024	Challenges in SimulST with a focus on low-latency processing of continuous, unsegmented speech.
3	LLaST: Improved End-to-end Speech Translation System Leveraged by Large Language Models	2024	Integration of large language models to enhance multilingual end-to-end speech translation.
4	Real-Time Transcription Language Translation Using WebRTC	2024	WebRTC-based real-time transcription and translation with enhanced NLP and speech processing techniques.
5	StreamSpeech: Simultaneous Speech-to-Speech Translation with Multi-task Learning	2024	Simultaneous speech-to-speech model using multi-task learning to learn policy and translation concurrently.
6	Recent Advances in End-to-End Simultaneous Speech Translation	2024	Survey of current research in SimulST with a focus on challenges and future directions.
7	Real Time Speech Transcription and Translation	2025	Use of APIs like Google Translate and gTTS for real-time translation combining TTS, MT, and ASR.

Table 2.1: Summary of Related Research Papers

CHAPTER 3

**SOFTWARE REQUIREMENTS
SPECIFICATION**

3.1 Assumptions and Dependencies

Assumptions

- **Stable Internet Connection:** Users are expected to have reliable, high-speed internet for optimal performance of real-time features.
- **Modern User Devices:** End-users will access the platform through devices equipped to handle advanced processing demands.
- **Familiarity with Video Conferencing Tools:** Users are assumed to be knowledgeable about common functionalities of video conferencing software.
- **High Server Availability:** Continuous uptime is expected for optimal user experience across different time zones.
- **User Adoption of Features:** Users are anticipated to embrace advanced functionalities that enhance their video conferencing experience.

Dependencies

- **WebRTC Framework:** The platform depends on WebRTC for real-time communication; any issues with it could affect service quality.
- **Third-Party AI Services:** Reliance on external APIs for features like translation may impact functionality based on their availability and performance.
- **Browser and Device Compatibility:** The performance hinges on support from modern browsers; compatibility issues could limit functionality.

3.2 Functional Requirements

3.2.1 System Feature 1: Real-Time Language Translation

Description: The platform must provide real-time language translation for spoken dialogue during video conferences, allowing users to communicate seamlessly in their native languages.

Requirements:

- Users can select their preferred languages before starting a meeting.
- The system should automatically transcribe spoken language and translate it into the selected language in real-time.
- Translated audio must be played back to users without significant latency.
- Display translated text captions on the screen to enhance understanding.

3.2.2 System Feature 2: Real-Time Chat Translation

Description:

The platform must support real-time translation of text messages exchanged in the chat during video conferences, ensuring clear communication between users speaking different languages.

Requirements:

- Users can send messages in their native language.
- The system should automatically detect the language of the message and translate it into the recipient's preferred language.
- Translation should occur with minimal delay to maintain the flow of conversation.

3.3 External Interface Requirements

3.3.0.1 User Interfaces

Feature	Description
Web Application Interface	Intuitive and easy-to-navigate UI for joining, hosting, and managing multilingual meetings.
Video Call Dashboard	Central hub showing video streams, chat, translated captions, and controls like screen share, download recording .
Language Selector	Allows users to choose their preferred input and output language for translation.
Chat Interface	Real-time translated text chat with the support of five different languages
Screen Sharing View	Displays shared content alongside active video feeds, optimized for clarity.
Responsive Layout	Compatible across devices and screen sizes using Tailwind CSS.

Table 3.1: User Interface Features of the Application

3.3.1 Hardware Interfaces

Hardware Component	Description
Microphone	Required for capturing user audio input for real-time speech recognition and translation.
Speakers / Headphones	Necessary for outputting the translated or original audio to the user.
Camera (Optional)	Enables video communication between users; enhances the conferencing experience.
Internet Connection	Stable and high-speed internet is essential for seamless audio/video streaming and translation services.
Device (PC / Smartphone / Tablet)	Any modern device capable of running a web browser or the mobile-responsive web application.

Table 3.2: Hardware Interface Requirements

3.3.2 Software Interfaces

Component	Technology Used	Why Chosen?
Frontend	Next.js, TypeScript, TailwindCSS	Fast, scalable, and responsive UI development.
Backend	Node.js, tRPC, Prisma, Supabase (PostgreSQL)	Efficient and scalable server-side communication and database management.
Video & Audio Communication	LiveKit (WebRTC)	Provides low-latency, high-quality audio and video streaming.
Real-Time Messaging	Pusher (WebSockets)	Enables instant messaging and real-time chat updates.
Speech-to-Text	Deepgram API	AI-driven speech recognition with high accuracy for multiple languages.
Translation	Google Translate API	Supports translation across a wide range of languages with reliable performance.

Table 3.3: Software Interfaces and Technologies Used

3.3.3 Communication Interfaces

Interface Type	Technology/Protocol Used	Purpose
Audio/Video Communication	WebRTC (via LiveKit)	Enables real-time audio and video conferencing with low latency.
Real-Time Messaging	WebSockets (via Pusher)	Allows bi-directional communication for instant chat translation.
External API Calls	Deepgram API, Google Translate API	Used for speech recognition and multilingual translation respectively.

Table 3.4: Communication Interfaces

3.4 Nonfunctional Requirements

3.4.1 Performance Requirements

- **Latency:** The platform should maintain a latency of less than 200 milliseconds for real-time audio and video processing to ensure smooth communication.
- **Concurrent Users:** The system must support up to 100 concurrent users in a single meeting without degradation in performance.
- **Response Time:** The application should respond to user interactions within 2 seconds to provide a seamless experience.

3.4.2 Safety Requirements

- **Data Backup:** Regular backups of user data must be performed to prevent data loss.
- **User Safety:** Implement features such as user reporting and blocking to ensure a safe environment for all participants.
- **Emergency Protocols:** Establish procedures for handling technical failures, including automatic error reporting and fallback options for users.

3.4.3 Security Requirements

- **Data Encryption:** All data transmitted between users and the server must be encrypted using TLS/SSL to protect sensitive information.
- **Authentication:** Implement robust authentication mechanisms, such as OAuth2, to ensure only authorized users can access the platform.

3.4.4 Software Quality Attributes

Attribute	Description
Usability	Intuitive interface designed for all users, including support for accessibility features.
Reliability	High accuracy in speech-to-text and translation through AI-powered APIs like Deepgram and Google Translate.
Scalability	Capable of handling up to 100 concurrent users with efficient load balancing and backend infrastructure.
Maintainability	Clean, modular codebase using modern frameworks (Next.js, tRPC, Prisma) for ease of updates and debugging.
Performance	Low-latency communication and quick processing enabled by WebRTC and WebSockets.
Portability	Web-based application accessible across devices and operating systems without dependency issues.

Table 3.5: Software Quality Attributes

3.5 System Requirements

3.5.1 Database Requirements

Requirement	Description
Database System	Supabase (PostgreSQL) chosen for scalability, ease of integration with Prisma ORM, and real-time capabilities.
User Management	Store user credentials, profile information, preferences, and session data securely.
Chat History	Maintain multilingual chat messages and timestamps for each session.
Translation Logs	Store original speech/text and translated outputs for review or feedback.
Backup	Regular automated backups to ensure data integrity and availability.

Table 3.6: Database Requirements

3.5.2 Software Requirements (Platform Choice)

Component	Technology Used and Reason
Frontend	Next.js, TypeScript, TailwindCSS – Fast, responsive UI development.
Backend	Node.js, tRPC, Prisma – Efficient APIs and scalable server-side logic.
Database	Supabase (PostgreSQL) – Real-time data syncing with relational storage.
Video/Audio	LiveKit (WebRTC) – Low-latency video and audio communication.
Messaging	Pusher (WebSockets) – Instant real-time chat updates.
Speech-to-Text	Deepgram API – High-accuracy AI transcription.
Translation	Google Translate API – Supports a wide range of languages.

Table 3.7: Major Software Requirements and Platform Choices

Component	Recommended Hardware Specifications
Client Device	• Processor: Dual-core 2.0 GHz or higher • RAM: Minimum 4 GB • Internet: Stable 2 Mbps or more • Webcam & Microphone for video/audio communication

Table 3.8: Hardware Requirements

3.6 Analysis Models: Agile Methodology to be Applied

The Agile methodology will guide the development of the project "The Bridging Language Barriers: The Role of AI in Real-Time Multilingual Translation for Video Conferencing." Agile emphasizes iterative progress, flexibility, and continuous feedback through the following key practices:

- **Iterative Development:** Break the project into smaller, manageable increments to deliver functional software regularly.
- **Collaboration:** Promote close collaboration among team members and stakeholders to ensure constant alignment with user needs.
- **Adaptive Planning:** Update plans and requirements based on real-time feedback and evolving project scope.
- **Continuous Testing:** Test throughout each iteration to identify and resolve defects early and frequently.
- **Delivery of Working Software:** Focus on delivering functional software at the end of every iteration, ensuring incremental progress.
- **Customer Feedback and Improvement:** Incorporate user feedback continuously to refine and enhance the system in subsequent iterations.

This flexible approach allows for rapid adaptation to changes and fosters continuous improvement throughout the project lifecycle.

CHAPTER 4

SYSTEM DESIGN

4.1 System Architecture

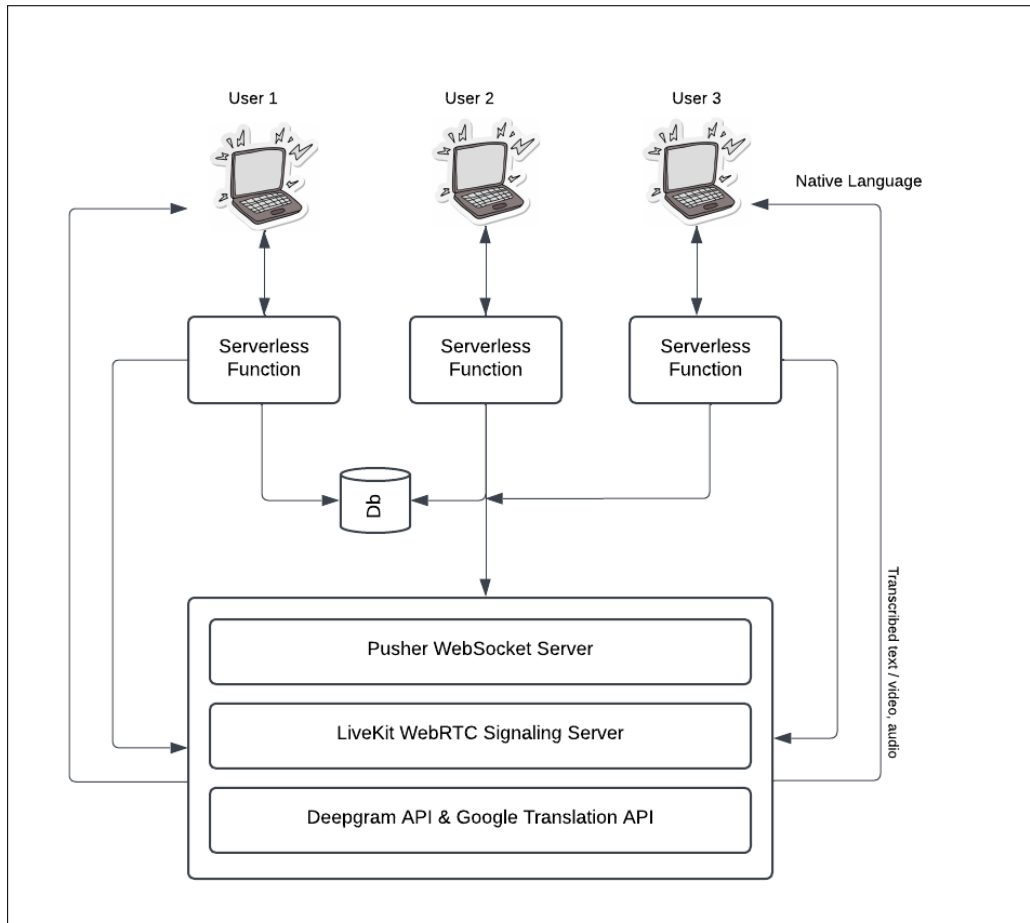


Figure 4.1: System Architecture Diagram

4.2 Data Flow Diagrams

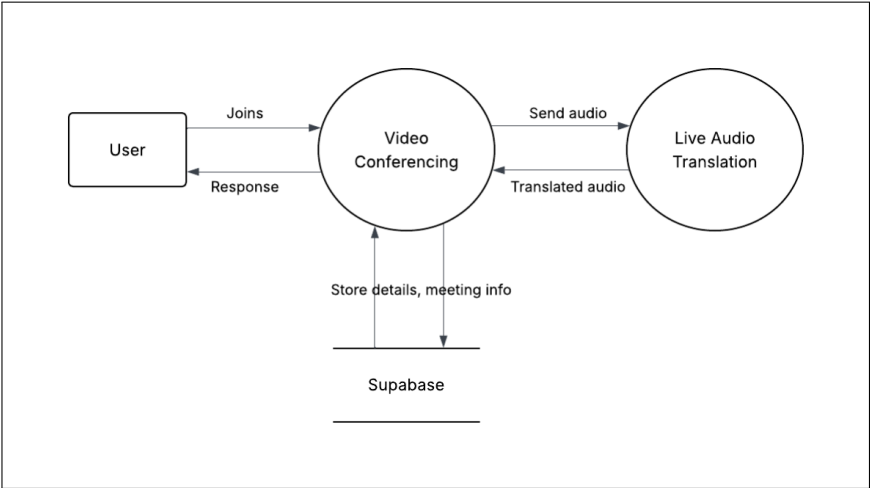


Figure 4.2: DFD 0

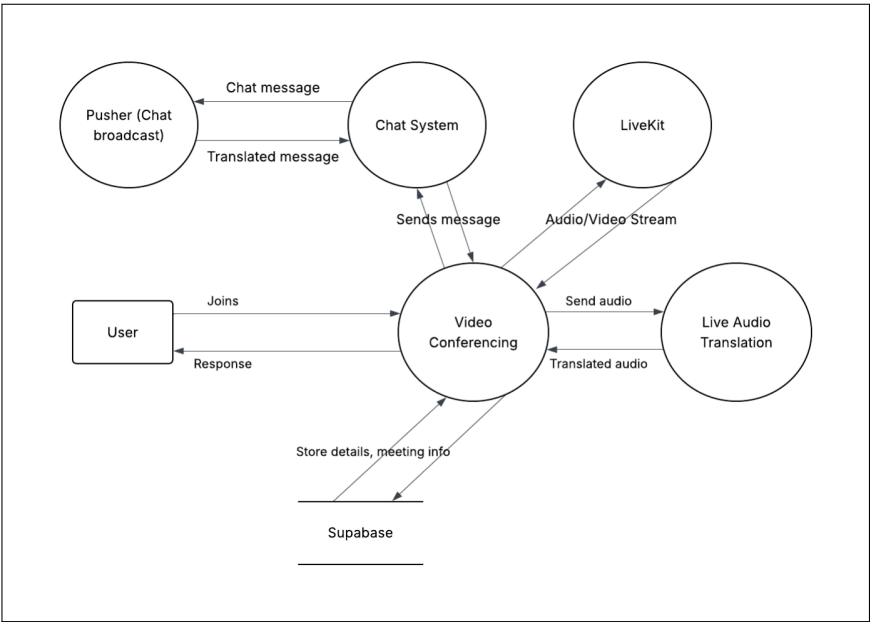


Figure 4.3: DFD 1

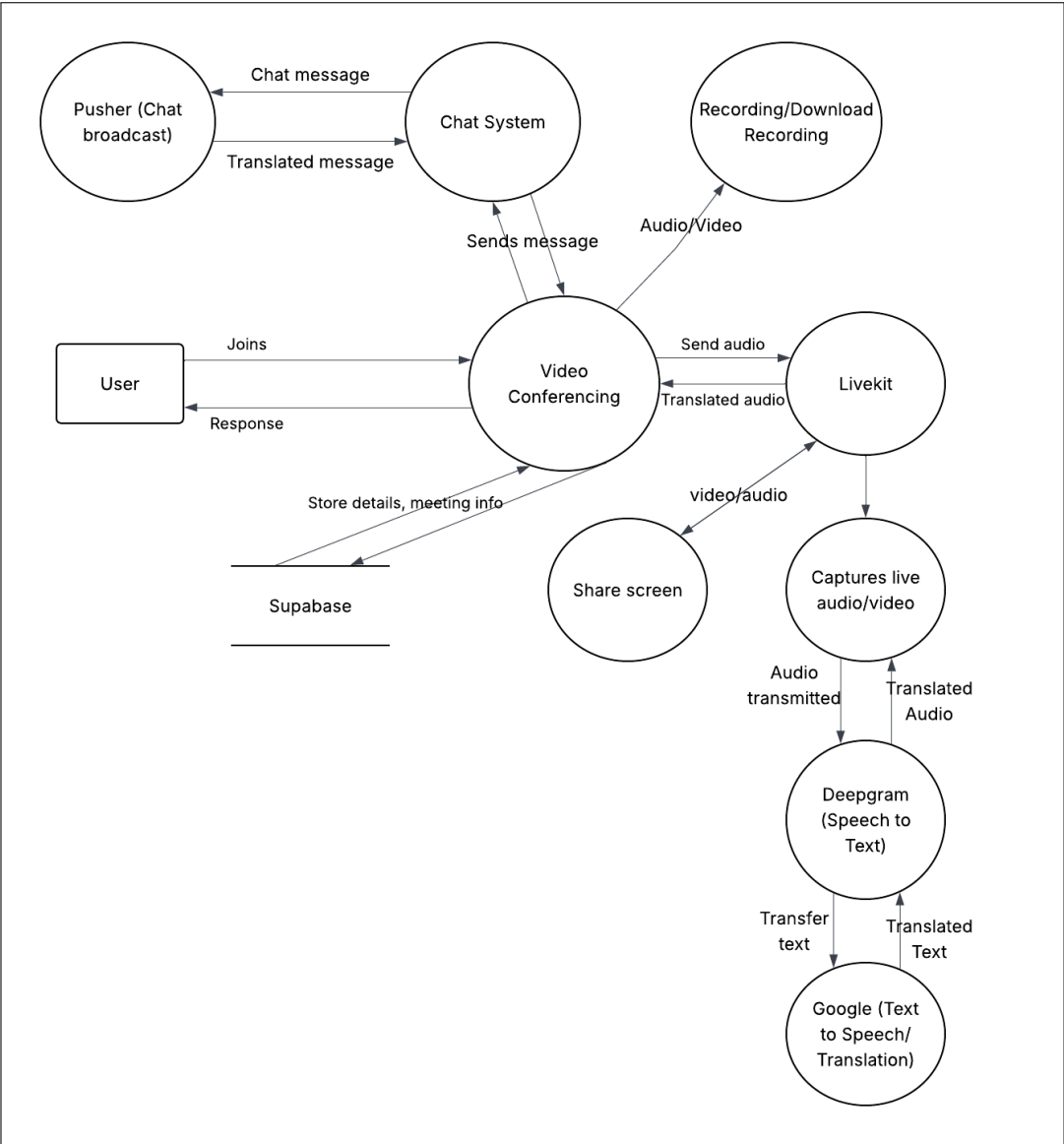


Figure 4.4: DFD 2

4.3 Entity Relationship Diagram

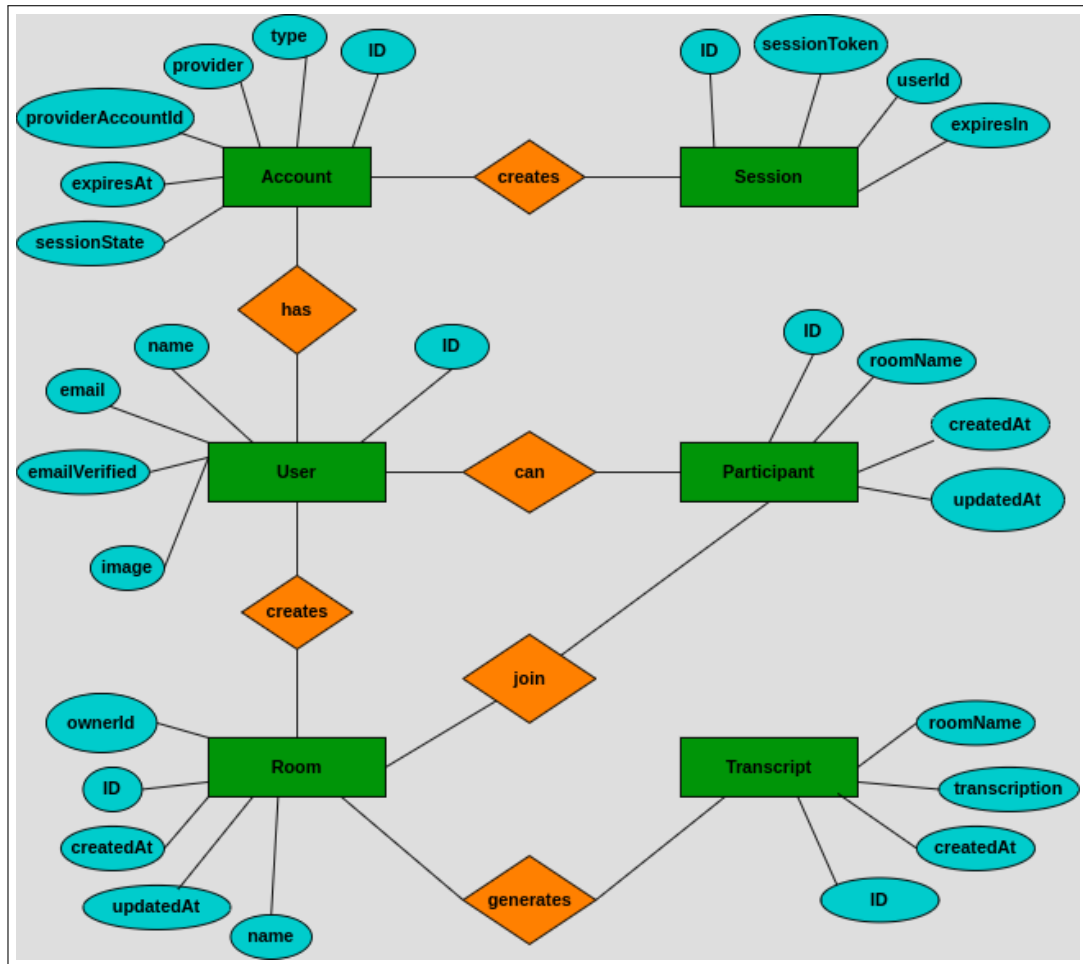


Figure 4.5: Entity Relationship Diagram

4.4 UML Diagrams : Class Diagram

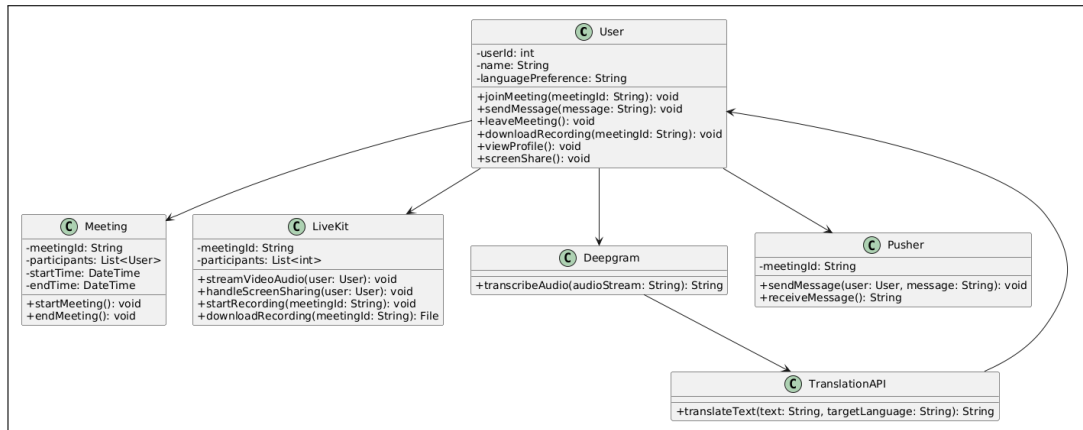


Figure 4.6: Class Diagram

4.5 UML Diagrams : State Diagram

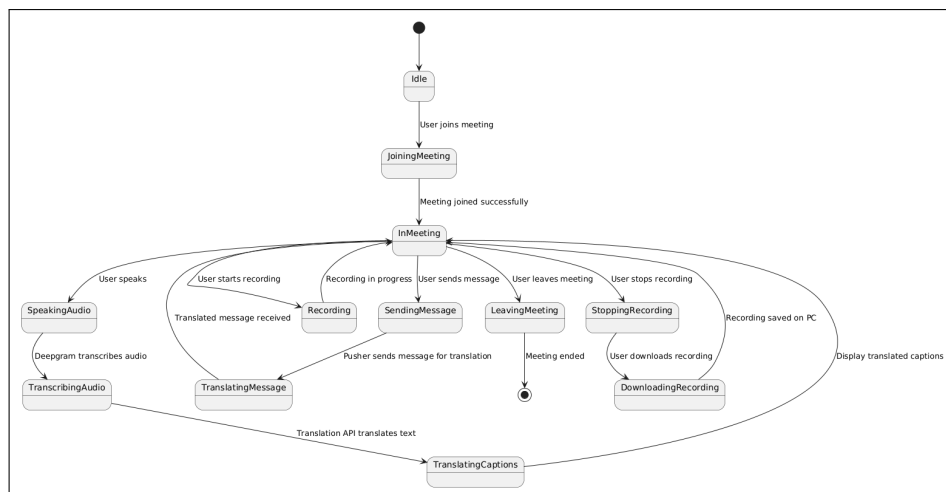


Figure 4.7: State Diagram

4.6 UML Diagrams : Activity Diagram

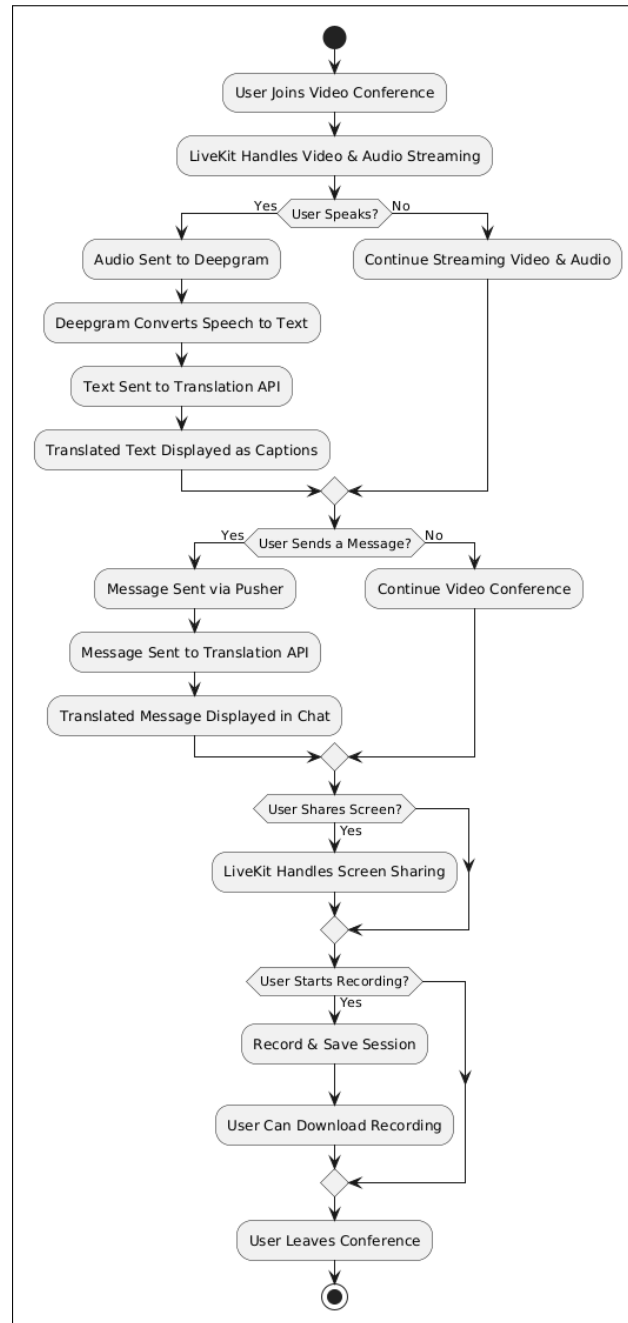


Figure 4.8: Activity Diagram

4.7 UML Diagrams : Use Case Diagram

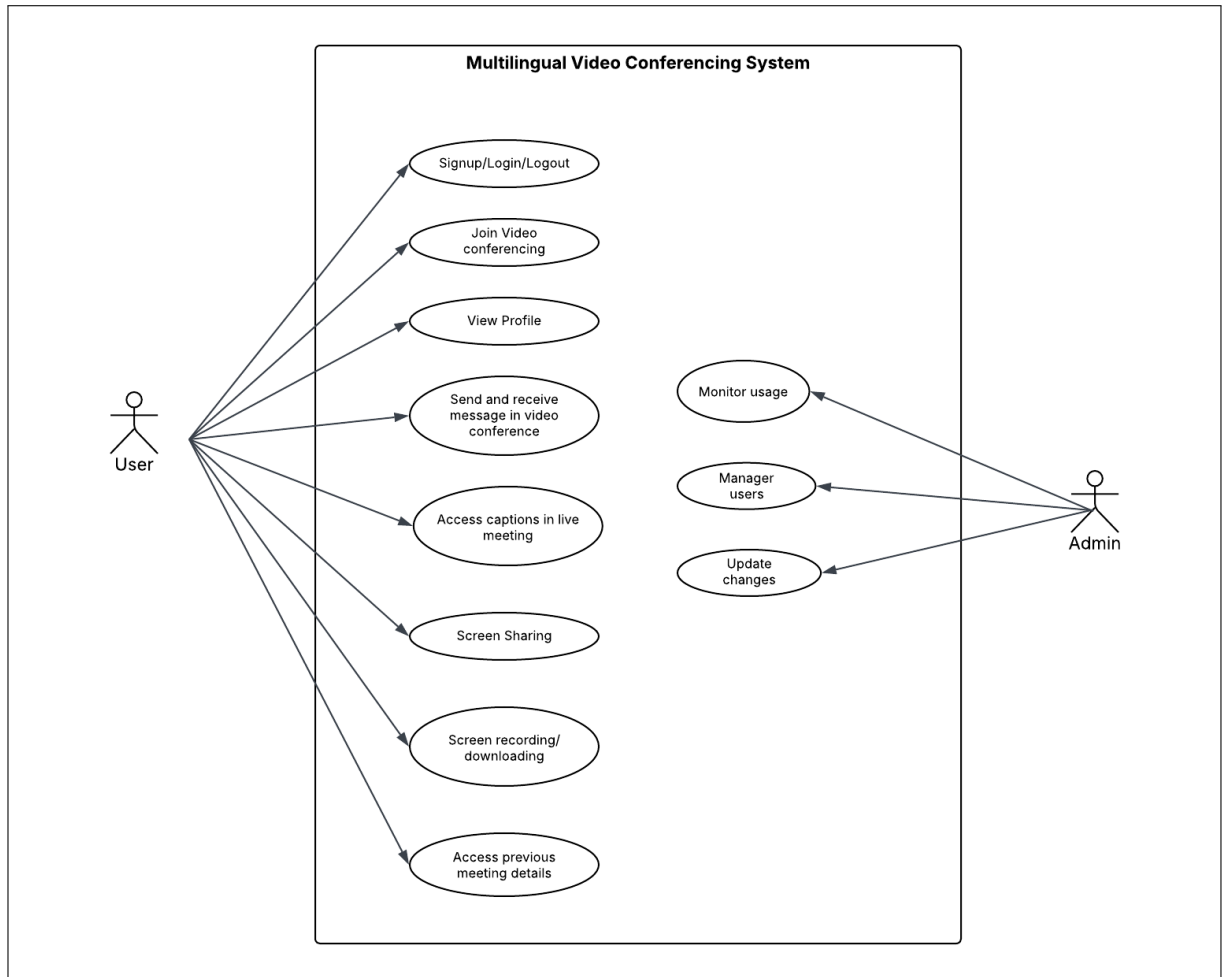


Figure 4.9: Use Case Diagram

4.8 UML Diagrams : Sequence Diagram

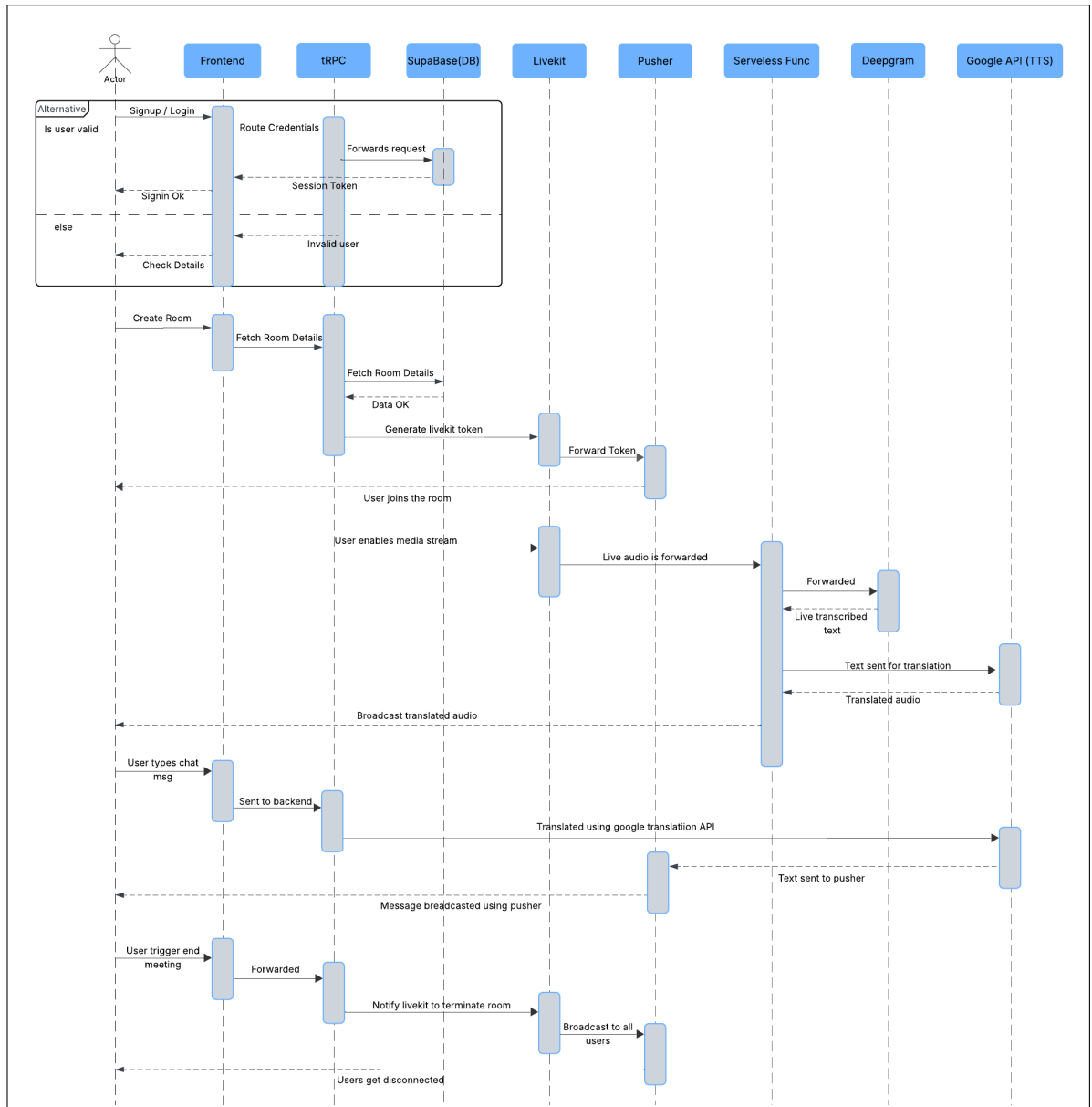


Figure 4.10: Sequence Diagram

CHAPTER 5

PROJECT PLAN

5.1 Project Estimate

5.1.1 Reconciled Estimates

- **Time Estimation:** Estimated completion time for the full system is approximately 8–10 weeks, including development, testing, and deployment phases.
- **Effort Estimation:** Around 300–350 person-hours expected based on tasks such as frontend/backend development, integration of APIs, and UI/UX design.
- **Cost Estimation:** Using mostly free-tier or open-source services (e.g., Supabase, Deepgram’s trial, LiveKit’s dev plan), the system can be built under a minimal budget; future scalability may incur additional costs.
- **Resource Allocation:** 3 team members with roles divided across frontend, backend, and integration to streamline development and reduce timeline overlaps.

5.1.2 Project Resources

- **Human Resources:** 3 developers collaborating on frontend, backend, and integration.
- **Technological Resources:** Laptops with internet access, GitHub for version control, and VS Code as the primary IDE.
- **Cloud Services:** Supabase for database hosting, LiveKit for video/audio, and Pusher for real-time messaging.
- **APIs and Tools:** Deepgram for speech-to-text, Google Translate API for translation, and WebSpeech API for TTS.

5.2 Risk Management

- **Data Privacy Risk:** Sensitive user data such as audio and chat logs must be protected. Mitigated by using secure APIs and database encryption.
- **Latency Issues:** Real-time translation and video communication can suffer from network delays. Mitigated by using low-latency tools like LiveKit and optimizing data flow.
- **API Limitations:** Dependency on third-party APIs (Deepgram, Google Translate) might cause service disruptions. Mitigated by implementing fallback mechanisms and monitoring usage quotas.
- **Scalability Risk:** As user load increases, performance may degrade. Addressed using Supabase's scalable backend and load testing.
- **Browser Compatibility:** Real-time features like WebRTC might not work equally across all browsers. Resolved through cross-browser testing and fallback options.

5.2.1 Risk Identification

- **Data Security Breach:** Unauthorized access to sensitive user data.
- **API Downtime:** External services like Deepgram or Google Translate may be unavailable.
- **Network Instability:** Poor internet connection may disrupt real-time features.
- **Scalability Challenges:** System may underperform under high user load.
- **Integration Bugs:** Issues during interaction of various system modules.

5.2.2 Risk Analysis

- **Data Security Breach:** High impact and high probability due to user data handling—requires encryption and secure authentication.
- **API Downtime:** Medium impact but moderate probability—needs fallback mechanisms or alternative APIs.
- **Network Instability:** High impact for real-time interactions—implement buffering or reconnection strategies.
- **Scalability Challenges:** High impact as users increase—requires load testing and autoscaling infrastructure.
- **Integration Bugs:** Moderate impact but frequent during development—use thorough testing and modular design.

5.2.3 Overview of Risk Mitigation, Monitoring, Management

- **Mitigation:** Implement secure coding practices, backup APIs, and robust testing strategies to reduce the likelihood of failures.
- **Monitoring:** Use logging, alerting tools, and real-time dashboards to track system performance and detect issues early.
- **Management:** Maintain a risk register, regularly assess threat levels, and update response strategies throughout the project lifecycle.

5.3 Project Schedule

5.3.1 Project Task Set

- **Requirement Analysis:** Gather functional and non-functional requirements from stakeholders.
- **System Design:** Design system architecture including frontend, backend, and database components.
- **Implementation:** Develop core modules such as video conferencing, speech-to-text, and translation.
- **Integration:** Connect all components including APIs, WebRTC, and real-time messaging.
- **Testing and Debugging:** Perform unit, integration, and user acceptance testing.
- **Deployment:** Host the application using Vercel for frontend and Supabase for backend.
- **Documentation:** Prepare user manuals, API documentation, and technical reports.
- **Maintenance:** Monitor system post-deployment and apply updates or patches as needed.

5.3.2 Task Network

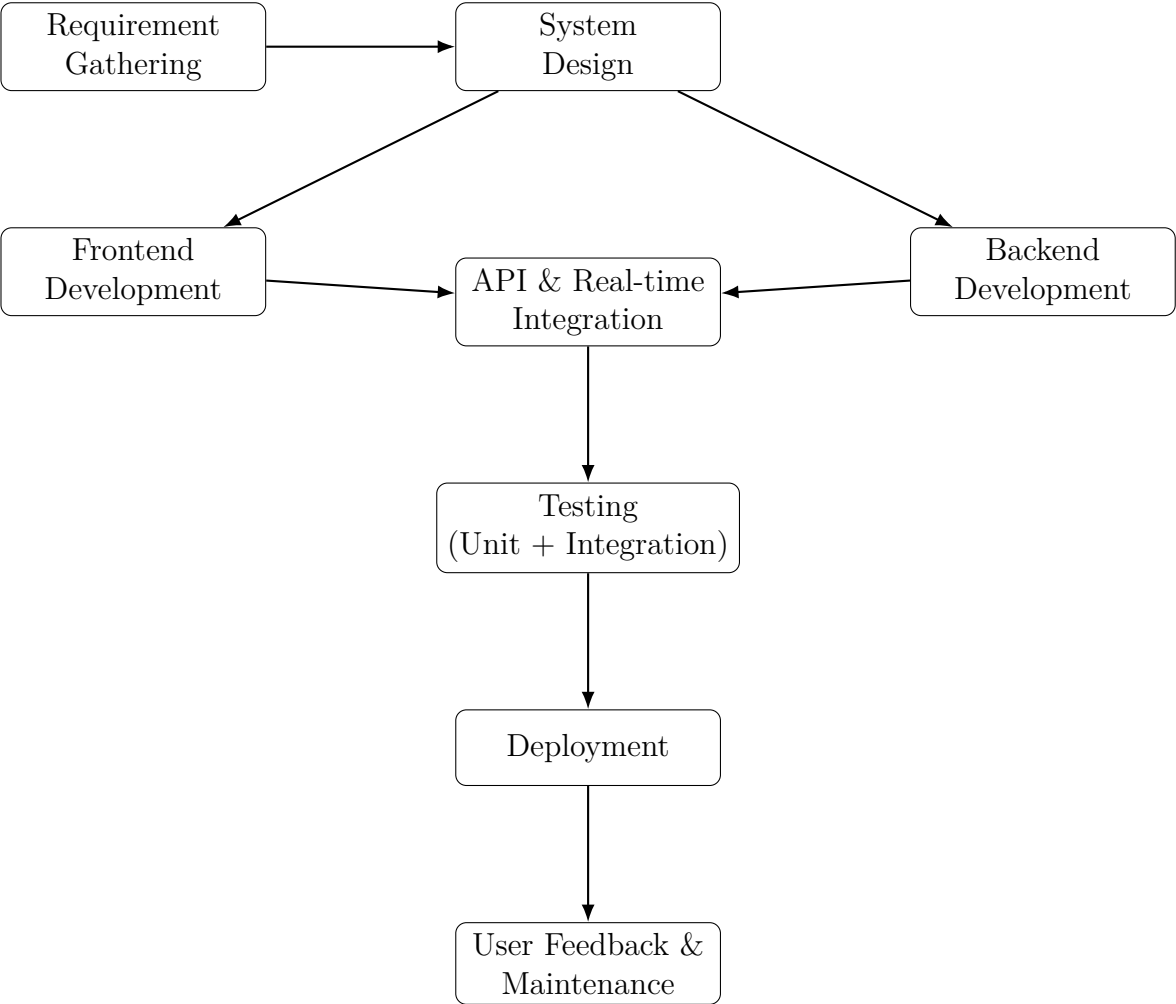


Figure 5.1: Task Network Diagram

5.3.3 Timeline Chart

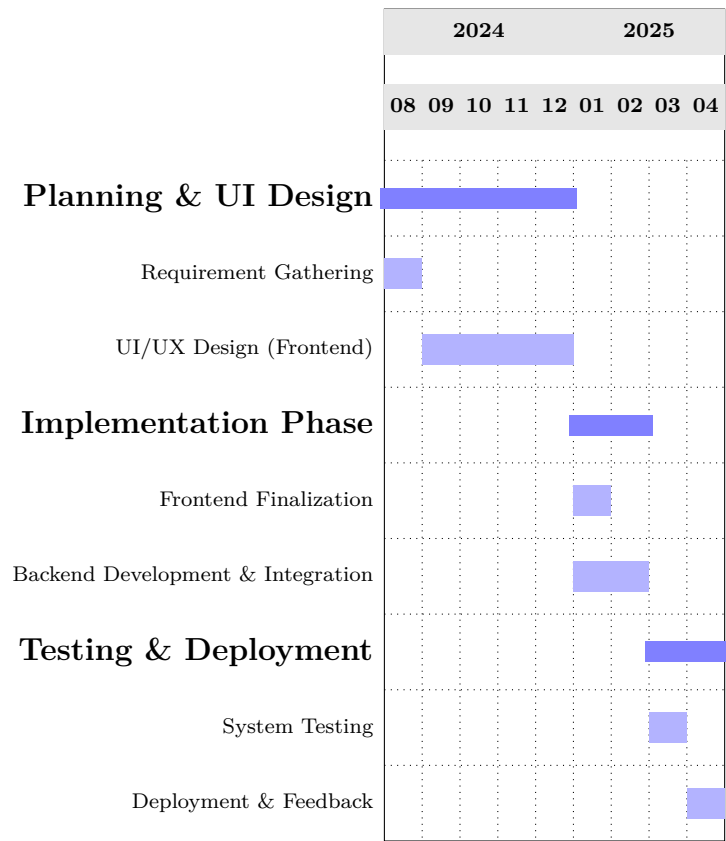


Figure 5.2: Timeline Chart

5.4 Team Organization

5.4.1 Team Structure

- **Team Size:** 3 Members
- **Role Distribution:**
 - **Afrin Shaikh:** Oversaw overall project development, managed collaboration, and ensured timely delivery of all phases.
 - **Prachi Said:** Designed and implemented the user interface using Next.js, TailwindCSS, and TypeScript. Also responsible for identifying appropriate APIs and actively contributing to literature review and research paper preparation.

- **Pratiksha Patare:** Developed the backend architecture using tRPC, Prisma, and Supabase (PostgreSQL). Took full responsibility for the entire project documentation, including technical specifications, reports, and system descriptions.
- **Collaboration Tools:** GitHub (version control), Figma (design prototyping), Google Meet (team meetings), Overleaf (LaTeX documentation).

5.4.2 Management Reporting and Communication

- Regular virtual meetings were held weekly using Google Meet to track progress, assign tasks, and address blockers..
- Team members communicated through a dedicated WhatsApp group and Google Chat for quick coordination.
- Major milestones and decisions were documented collaboratively using Google Docs and shared with all team members.
- Final reports and presentations were prepared and refined through Overleaf to maintain consistency and quality.

CHAPTER 6

PROJECT IMPLEMENTATION

6.1 Overview of Project Modules

LiveKit (Real-time Audio and Video Streaming)

Livekit is a Live Streaming Platform that uses real-time WEBRTC Technology to facilitate seamless audio and video communication among multiple users. It utilizes WEBRTC (Web Real-Time Communication) to establish peer-to-peer connections and effectively transmit audio, video, and data streams.

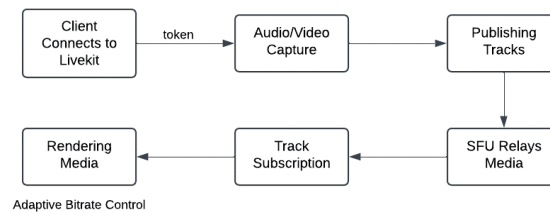


Figure 6.1: LiveKit Workflow

LiveKit Workflow :

1. **Establishing connection :** A user opens the client application (browser or mobile app) and then the client requests a connection token from the backend to join a LiveKit room. The backend generates an access token with a unique identity using a LiveKit API key and secret. This token includes permissions such as **canPublish**, **canSubscribe**, and **canPublishData**. Then the token is sent back to the client. Client connects to the LiveKit server using WebRTC. ICE (Interactive Connectivity Establishment) protocol is used to negotiate and establish a connection between peers.
2. **Media Capture and Publishing :** Browser APIs capture audio and video using **getUserMedia()**. Media streams are encoded and prepared for transmission. Local audio and video tracks are created using LiveKit's **createLocalTracks()** API. Tracks are attached to a LocalParticipant object, representing the user. Media tracks are published to the LiveKit room using **room.localParticipant.publishTrack()**. LiveKit sends the audio and video streams to the SFU (Selective Forwarding Unit) server.

3. **Establishing WebRTC connection :**

ICE Negotiation - ICE protocol initiates the negotiation to establish a connection. STUN (Session Traversal Utilities for NAT) servers identify public IPs and assist in peer connection. TURN (Traversal Using Relays around NAT) servers relay media traffic if direct peer-to-peer communication fails.

SDP (Session Description Protocol) Exchange - Offer and answer messages containing media configuration (codecs, encryption, etc.) are exchanged. Once both peers agree on media configuration, the connection is established.

4. **Media Transmission and Routing :**

SFU (Selective Forwarding Unit) - LiveKit uses an SFU architecture where media streams from one participant are forwarded to others. SFU optimizes bandwidth by relaying only the required streams to each participant. Media is routed through LiveKit's SFU, ensuring scalability with multiple participants. Media streams are encoded using codecs such as VP8 (video) and Opus (audio).

Track Subscription - Other participants subscribe to the published audio and video tracks. Subscribed tracks are rendered on the respective participant's UI.

5. **Data Channel and Chat Support :** WebRTC's Data Channel is used for real-time exchange of text, messages, and other non-media data. Participants can send and receive text messages alongside audio and video.

6. **Bandwidth Management and Adaptive Bitrate Control :** LiveKit dynamically adjusts media quality to adapt to changing network conditions. If network bandwidth decreases, video quality is reduced to maintain smooth communication. Multiple versions of the same media stream are sent at different quality levels. SFU automatically switches between streams based on the receiving client's bandwidth.

Deepgram (Speech to text processing)

Deepgram is an AI-powered speech recognition API that converts audio into text in real time. It uses end-to-end deep learning models trained on vast datasets to ensure high accuracy, even in noisy environments or with different accents. Here's a detailed breakdown of how Deepgram processes and transcribes speech effectively:

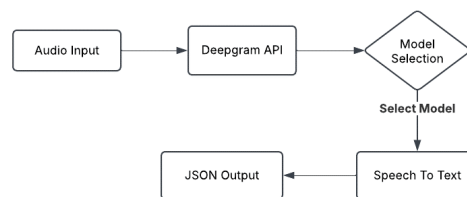


Figure 6.2: Deepgram (Speech to text processing)

Deepgram Workflow :

1. **Audio Input and Streaming Setup :** The audio is captured in real time from the user's microphone during a video call. **LiveKit** handles the audio/video streams and transmits the audio in real time to Deepgram using WebRTC. The audio is transmitted over **WebSockets** to Deepgram's API endpoint (`wss://api.deepgram.com/v1/listen`) for real-time transcription. WebSocket-based communication allows bi-directional, low-latency data flow.
2. **Audio Ingestion and Preprocessing :** Deepgram receives the audio stream in formats like PCM, WAV, or Opus. The system performs **signal preprocessing** to filter out background noise, normalize audio levels, and enhance speech clarity. Deepgram supports multiple audio channels, making it capable of identifying and processing individual speakers in a conversation.
3. **Speech-to-Text Conversion Using Deep Learning Models :** Deepgram uses convolutional neural networks (CNNs) and recurrent neural networks (RNNs) to analyze audio signals and extract relevant speech features. These models perform **phoneme-level analysis** and convert audio signals into linguistic representations. The system matches these patterns to produce high-accuracy transcriptions in real time. As audio data is streamed, Deepgram continuously updates the

transcription with **partial results** that improve as the model processes more data. Deepgram's latency is as low as **300 milliseconds**, making it one of the fastest speech-to-text solutions available.

4. **Post-Processing and Output :** Deepgram applies additional post-processing techniques to improve transcription quality, including punctuation and capitalization. Final transcriptions are returned as structured JSON responses, which can be processed further for translation using the Google Translate API.

Deepgram's use of WebSockets ensures minimal overhead, enabling low-latency delivery of transcription results. WebSocket-based data streaming reduces the handshake delay common in HTTP requests. Deepgram's GPU-accelerated models process and transcribe audio in real time, allowing for seamless live transcription.

Multilingual Translation (Google Translate API)

The transcribed text is sent to the Google Translate API using an HTTP POST or GET request.

Request Format :

```
POST https://translation.googleapis.com/language/translate/v2
Body:
{
  "q": "Hello, how are you?",
  "target": "hi", // Target language code (e.g., 'hi' for Hindi)
  "source": "en", // Source language (op, auto-detect if not provided)
  "format": "text"
}
```

Response Format :

```
{
  "data": {
    "translations": [
      {
        "translatedText": "नमस्ते, आप कैसे हैं?",
        "detectedSourceLanguage": "en"
      }
    ]
  }
}
```

Google Translate API Workflow :

1. **Language Detection** - If no source language is specified, the API auto-detects the language. Language detection uses statistical models and pattern analysis.
2. **Translation Model** - Applies Neural Machine Translation (NMT) models. Context-aware translation improves accuracy by understanding surrounding text.
3. **Output Generation** - Returns translated text in the target language.

Text To Speech (TTS) Conversion : The translated text is sent to a TTS engine. Converts translated text into natural-sounding speech. Supports multiple languages and voice types. Response in audio format (MP3, WAV, etc.).

Real time messaging and event handling (Pusher API)

Pusher is used for real-time event-driven communication. It ensures that when a message is sent, it is instantly delivered to all connected clients. In this project, it is primarily used for: Real time chat updates, Synchronizing translated messages, Broadcasting messages to all participants in a chat.

User Sends Message - User types and sends the message in their native language.

Message Payload :

```
{
  "user": "Alice",
  "message": "Hello, how are you?",
  "language": "en",
  "targetLanguage": "hi"
}
```

The message is sent to the backend via an HTTP API or WebSocket.

Message is processed and translated - Backend receives the message. The original text is sent to Deepgram (if audio) or passed directly. Transcribed or original text is then sent to the Google Translate API. The translated message is generated in the target language.

Message sent to pusher - The translated message is sent to Pusher. Backend triggers a Pusher event:

```
pusher.trigger("chat-channel", "new-message", {  
  user: "Alice",  
  message: "नमस्ते, आप कैसे हैं?",  
  originalMessage: "Hello, how are you?",  
  language: "hi",  
  originalLanguage: "en"  
});
```

Channel - chat-channel

Pusher Broadcasts the Message - Pusher pushes the message to all connected clients. Clients subscribed to chat-channel receive the message instantly. Real-time update in the chat interface.

Client receives message and updates UI - All connected clients receive the translated message. WebSocket listener updates the UI with the translated text:

```
pusher.subscribe("chat-channel").bind("new-message", (data) => {  
  displayMessage(data.user, data.message, data.language);  
});
```

tRPC (API communication)

tRPC (TypeScript Remote Procedure Call) is used to enable seamless, type-safe communication between the frontend and backend without the need for REST API. It eliminates the need for writing multiple endpoints, handling API contracts, or manually defining request/response formats. The main goal is to ensure that the types defined on the backend automatically sync with the frontend, reducing the risk of errors and speeding up development.

Security and access control

- **User Authentication and Session Management :**

Users log in using credentials (email, password) or OAuth (Google, etc.), verified against a secure database.

After login, a session token (JWT) is issued and stored in cookies or local storage to maintain the user's session.

Tokens have an expiry time, and refresh tokens ensure prolonged sessions without frequent re-login.

- **Role-Based Access Control (RBAC) for Meetings :**
 1. **Participant:** Can join and interact with limited privileges.
 2. **Guest:** May have view-only access.

6.2 Tools and Technologies Used

Category	Technology/Tool	Purpose
Frontend	Next.js, TypeScript, Tailwind-CSS	Building fast, responsive, and scalable user interfaces
Backend	Node.js, tRPC, Prisma ORM	API handling, type-safe backend logic, and database access
Database	Supabase (PostgreSQL)	Managed cloud database and storage
Real-Time Communication	LiveKit (WebRTC)	Audio-video streaming and conferencing
Real-Time Messaging	Pusher (WebSockets)	Instant chat and event broadcasting
Speech-to-Text	Deepgram API	Real-time voice transcription
Translation	Google Translate API	Multilingual translation of chat and audio
Authentication	Supabase Auth	User registration and login
Deployment	Vercel	Hosting frontend and back-end
Version Control	Git, GitHub	Source code management and collaboration

Table 6.1: Tools and Technologies Used

6.3 Algorithm Details

6.3.1 Algorithm 1: Real-Time Audio Translation Workflow

1. **Audio Capture:**
 - User’s microphone captures voice using ‘getUserMedia()’ API.

2. Speech Recognition (Transcription):

- Audio is streamed to Deepgram API.
- Deepgram processes the audio and returns real-time transcriptions.

3. Translation:

- The transcription is passed to the Google Translate API.
- The API returns the translated text in the desired target language.

4. Text-to-Speech :

- Translated text is converted to speech using Web Speech API.
- Audio is played for the recipient user in their preferred language.

5. Display Captions:

- Real-time translated captions are rendered on the user interface.

6.3.2 Algorithm 2: Real-Time Chat Translation Workflow

1. User Message Input:

- User types a message in their native language in the chat box.

2. Message Handling:

- The message is sent to the backend via WebSocket (Pusher).

3. Translation Process:

- The message is processed and translated into the receiver's language using the Google Translate API.

4. Message Broadcasting:

- Both original and translated messages are sent to the relevant users.

5. Display:

- Users see both the original and translated messages in the chat interface.

CHAPTER 7

SOFTWARE TESTING

7.1 Type of Testing

- **Unit Testing:** Tests individual components like UI elements, API endpoints, and utility functions.
- **Integration Testing:** Verifies the interactions between frontend, backend, and third-party APIs.
- **System Testing:** Validates the complete system workflow, ensuring all modules function as expected.
- **Acceptance Testing:** Ensures the final product meets the initial project requirements and user expectations.

7.2 Test cases & Test Results

Test Case ID	Description	Expected Result	Status
TC01	User joins meeting room	User is successfully connected to LiveKit room	Pass
TC02	Real-time audio translation	Audio is captured, translated, and played back in real-time	Pass
TC03	Chat translation	Messages are translated correctly before being displayed	Pass
TC04	Speech-to-text transcription	Spoken audio is converted to text with acceptable accuracy	Pass
TC05	Network failure handling	Reconnection is attempted and media streams resume	Pass
TC06	Screen sharing functionality	Screen is shared and visible to all participants	Pass
TC07	Mobile responsiveness	UI renders properly on both Android and iOS browsers	Pass
TC08	Unauthorized access attempt	User without token is denied entry to room	Pass
TC09	Audio translation delay	Translated audio should play within 1-2 seconds of original speech	Fail
TC10	Caption synchronization	Captions should appear in sync with spoken words	Fail
TC11	Screen sharing on mobile	Screen sharing should be supported on Android devices	Fail
TC12	Token expiration handling	Expired token should redirect to login with proper message	Fail
TC13	User not signed in	Unauthenticated user should be prompted to sign in before joining	Fail
TC14	No internet connectivity	User should see "No connection" message and retry option	Fail

Table 7.1: Test Cases and Test Results Including Fail Scenarios

CHAPTER 8

RESULTS

8.1 Outcomes

1. Successfully built a multilingual video conferencing system with real-time audio and chat translation.
2. Enabled users from different linguistic backgrounds to communicate seamlessly.
3. Implemented screen sharing, recording, and transcription features for enhanced collaboration.
4. Ensured scalable backend with efficient real-time data handling using LiveKit and Pusher.

8.2 Screen Shots

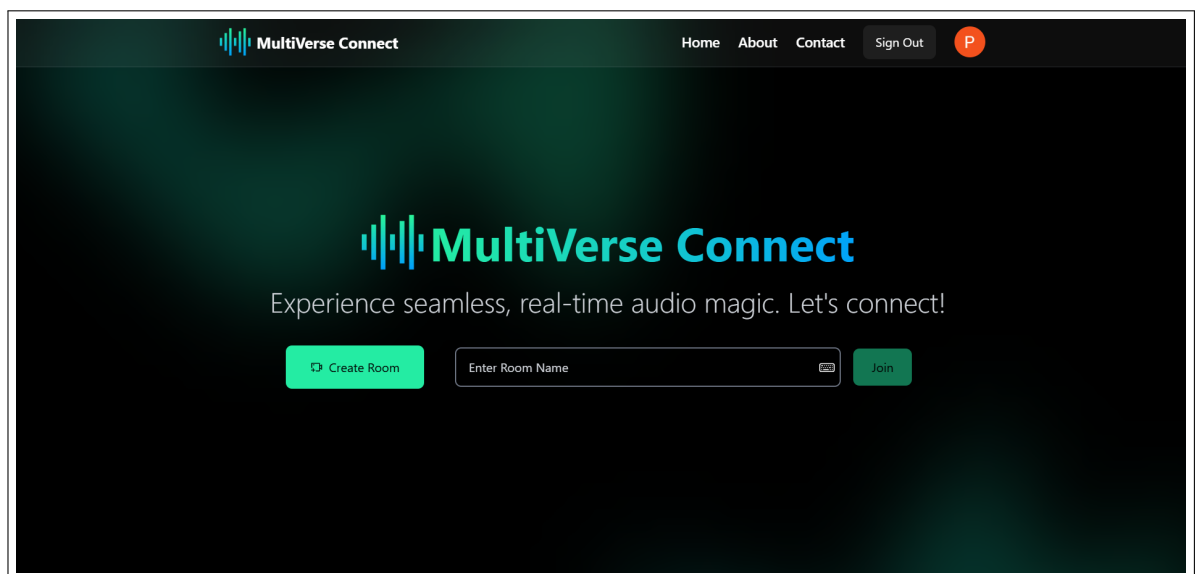


Figure 8.1: Home Page

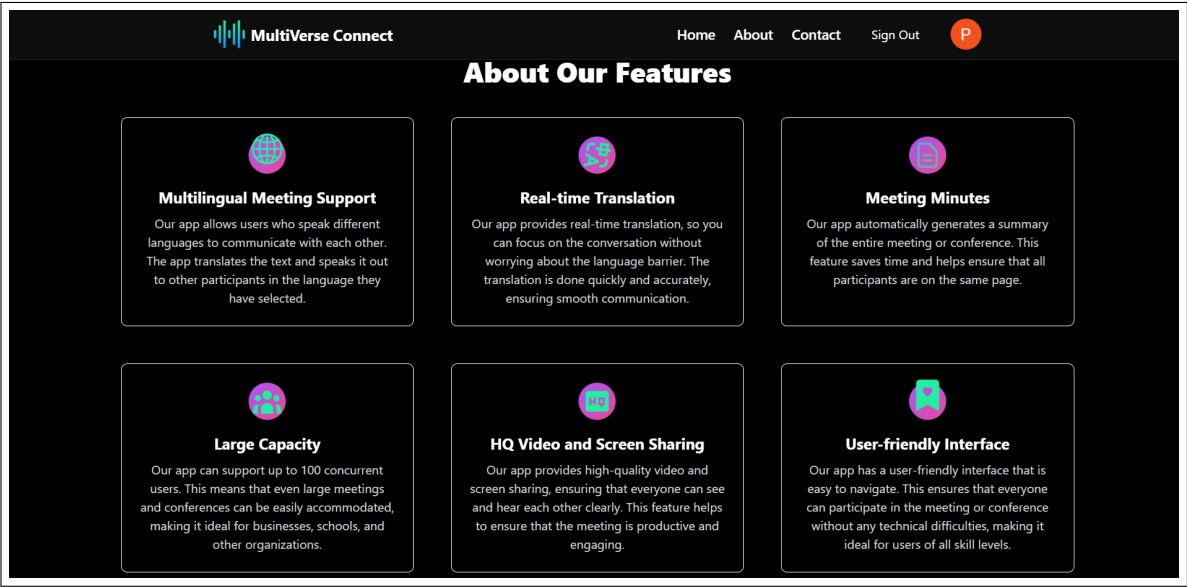


Figure 8.2: About Us Page

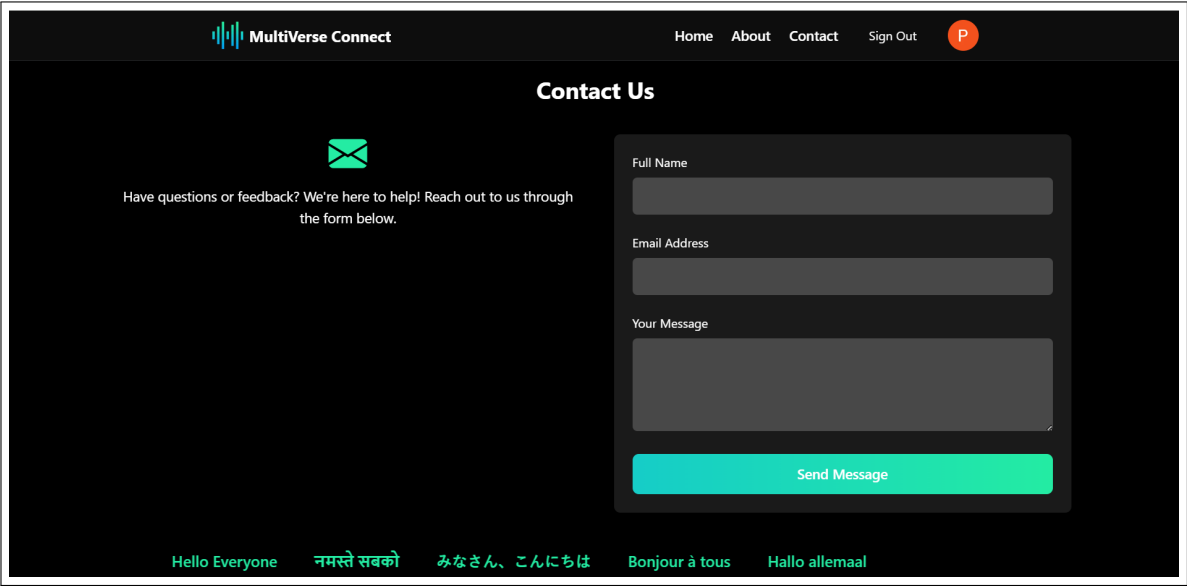


Figure 8.3: Contact Page

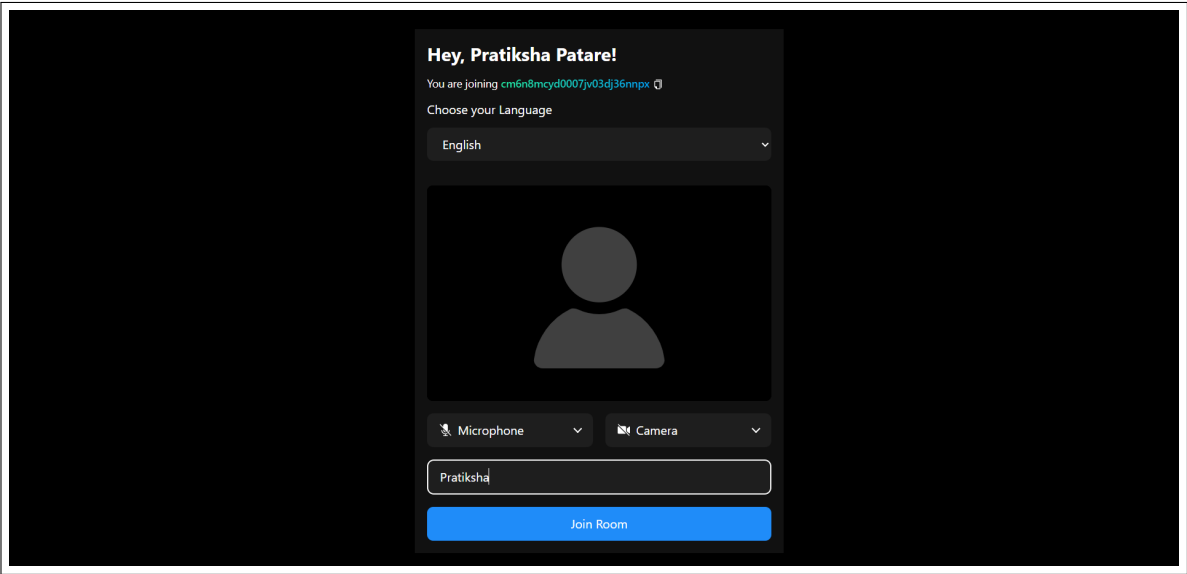


Figure 8.4: Join Meeting

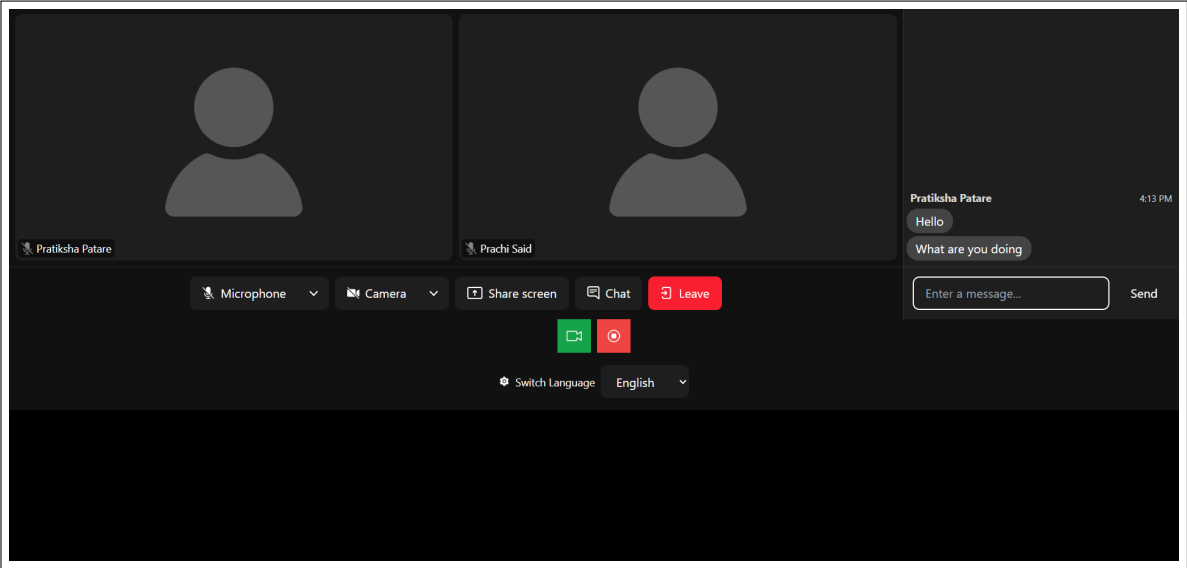


Figure 8.5: English Chat from Sender End

The Role of AI in Real-Time Multilingual Translation for Video Conferencing

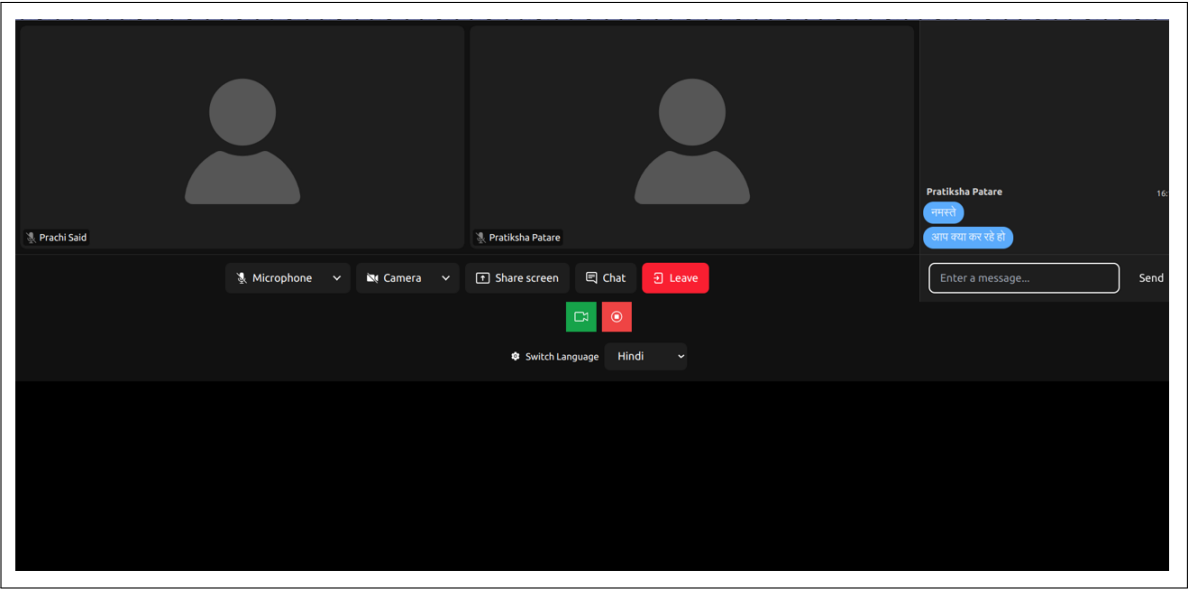


Figure 8.6: Hindi Chat Received at Receiver End

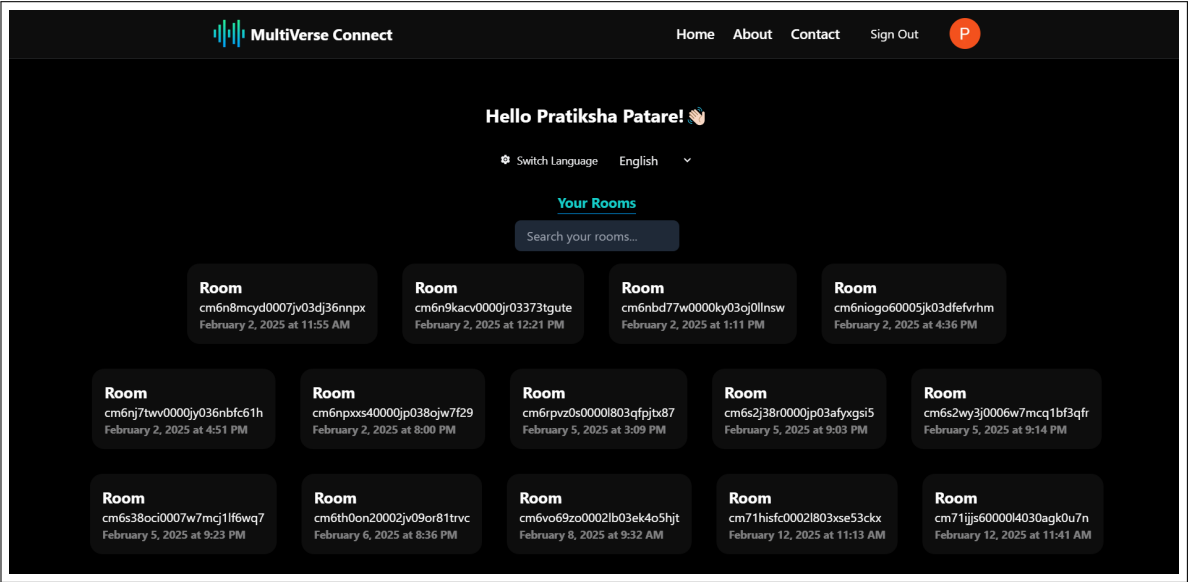


Figure 8.7: Profile Details

The Role of AI in Real-Time Multilingual Translation for Video Conferencing

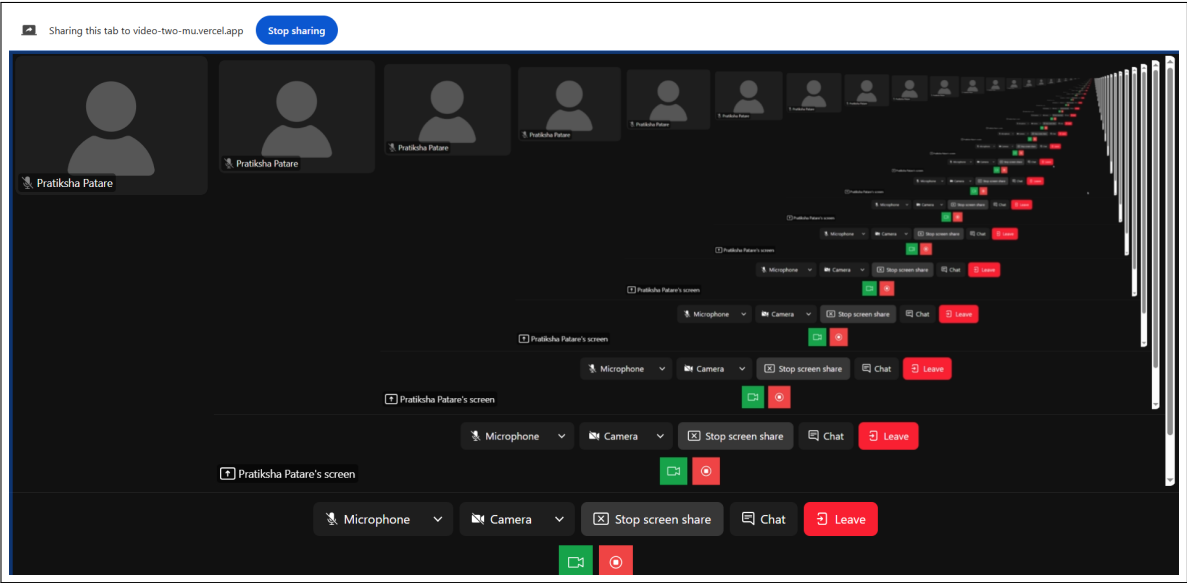


Figure 8.8: Screen Sharing

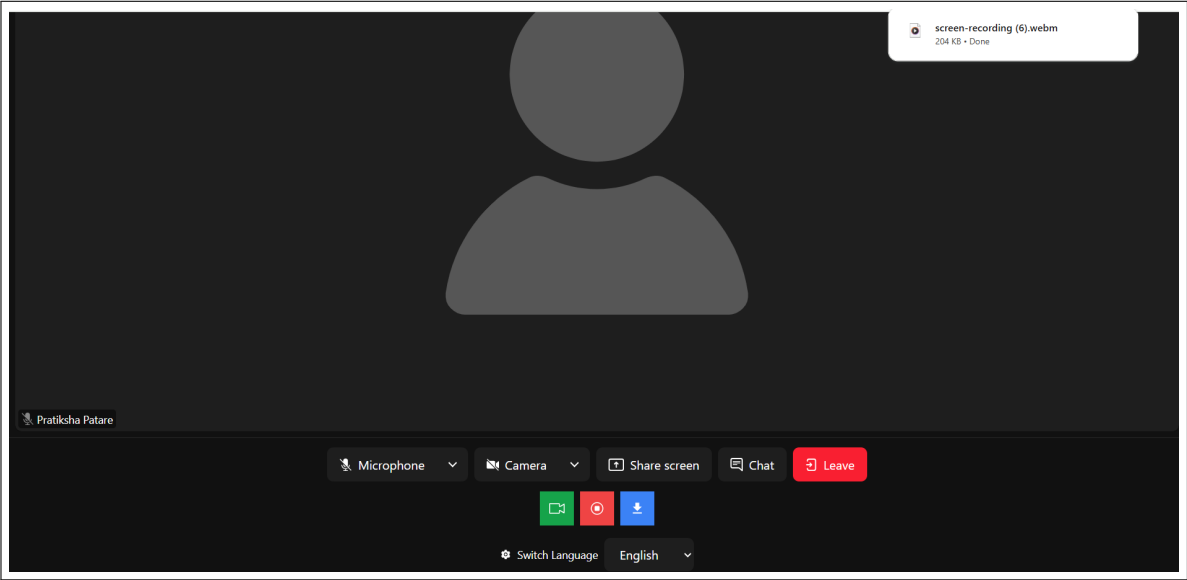


Figure 8.9: Download Recording

CHAPTER 9

CONCLUSIONS

9.1 Conclusions

The development of a real-time multilingual speech-to-speech translation and transcription system marks a significant advancement in overcoming language barriers in video conferencing. By integrating technologies such as Deepgram for speech recognition, Google Translate API for translation, and LiveKit for seamless audio-video streaming, this solution ensures low-latency and high-fidelity communication. The system's scalability and efficiency make it a viable option for multilingual collaboration, paving the way for future improvements in AI-driven communication platforms. With continued advancements, such technology can further enhance inclusivity and accessibility in global interactions.

9.2 Future Work

1. Integration of advanced AI translation models for improved accuracy in real-time communication.
2. Mobile application development for native iOS and Android platforms.
3. Adding support for more languages and regional dialects.
4. Improving noise cancellation and audio quality with ML-enhanced audio processing.
5. Enhancing security with end-to-end encryption for all data streams.
6. Incorporating emotion detection and sentiment analysis for better user insights.

9.3 Applications

- Startups and MNCs with International Work Culture
- Freelancers with Daily Foreign Communication
- Content Creators & Live Streamers
- Educational Organizations

CHAPTER 10

APPENDIX

10.1 Appendix A

Problem statement feasibility assessment using, satisfiability analysis and NP Hard, NP-Complete or P type using modern algebra and relevant mathematical models.

Component/Problem	Satisfiability Perspective	Complexity Class
Speech-to-Text Conversion	Feasible using Deep Learning models and ASR APIs	P (Polynomial Time)
Language Translation	Feasible using pretrained multilingual models	P (Polynomial Time)
Real-time Audio/Video Communication	Requires stable connectivity and low-latency protocols (e.g., WebRTC)	P
Resource Scheduling under Bandwidth Limits	Optimization problem under constraints	NP-Hard
Multi-user Chat Sync	Achievable using WebSocket-based protocols like Pusher	P
Adaptive Bitrate Streaming	Requires dynamic adjustment logic	P

Table 10.1: Feasibility and Complexity Classification of Project Components

10.2 Appendix B

Details of paper publication: name of the conference/journal, comments of reviewers, certificate, paper.

Sr. No.	Paper Title	Conference / Journal Name
1	Bridging Language Barriers: The Role of AI in Real-Time Multilingual Translation for Video Conferencing	International Journal of Innovative Research in Technology
2	Real-Time Multilingual Audio Translation and Transcription in Video Conferencing	5th International Conference on Engineering, Social-Sciences, And Humanities (ICESSU – 2025)

Sr. No.	Certificate Description
1	Certificate of Publication for the paper titled “Bridging Language Barriers: The Role of AI in Real-Time Multilingual Translation for Video Conferencing” in the International Journal of Innovative Research in Technology
2	Certificate of Presentation for the paper “Real-Time Multilingual Audio Translation and Transcription in Video Conferencing” at the 5th International Conference on Engineering, Social-Sciences, And Humanities (ICESSU – 2025)
3	Certificate of Participation in the event “Impetus and Concepts”
4	Certificate of Participation in the National Level Project Competition-2025 (NLPC-25), organized by the Institution Of Electronics and Telecommunication Engineers (IETE), Pune

Paper Acceptance Details

The image is a screenshot of an email interface. At the top, the subject line reads "Paper 168716 Acceptance Notification from IJIRT" with a small "Inbox x" indicator. The sender is "Editor in Chief <editor@ijirt.org>" and the recipient is "to me". The email body features a dark blue header with the text "Dear Pratiksha Palare, Welcome to IJIRT!". Below this, the text states: "Your manuscript has been **Accepted** for publication in the **International Journal of Innovative Research in Technology(IJIRT)**." It then provides the "Paper Id: 168716" and the "Paper Title: 'Bridging Language Barriers: The Role of AI in Real-Time Multilingual Translation for Video Conferencing'". The "Review Status" is marked as "Accepted" in yellow. The "Processing Charges" are listed as "1500" in yellow. A paragraph follows: "Kindly login with your Paper ID and email id to the following link to proceed further." Another paragraph states: "Hardcopy of Journal is available at 500 INR. Please visit the Login area to apply for the same." At the bottom of the email content is a dark blue button labeled "Pay Processing Charges". The email is dated "Mon, 21 Oct 2024, 08:43" and includes standard email icons for star, smile, reply, and more options.

Survey Paper Acceptance

Congratulations! Paper Accepted for Presentation at 5th ICESSU Conference

ICessu Conference <info@icessu.net>
to me ▾

Dear Pratiksha Patare,

We are delighted to inform you that your paper titled "**Real-Time Multilingual Audio Translation and Transcription in Video Conferencing**" has been accepted for presentation at the 5th ICESSU Conference. Congratulations on this significant achievement!

To confirm your participation, we kindly request you to complete the registration process. The total registration fee for the presentation is INR 3500. Upon successful registration, we will provide you with the final acceptance letter.

Registration Process:
To secure your slot at the conference, please complete your registration by making the payment using the following link: <https://www.ifero.in/payment/>

Registration Benefits:

- ✓ Proceedings Book publication (with ISBN: 978-81-976209-7-3) will be submitted to Scopus for Evaluation and publication
- ✓ Certificate of Presentation – Official recognition of your contribution
- ✓ CPD Certificate – Enhance your professional credentials
- ✓ 1-Year CPD Membership – Expand your academic and professional network

To complete the registration, follow these steps:

- After opening the link, enter the conference name: ICESSU
- Select the participation category:
- Select the currency: INR
- Enter the fee:

Mohammed Azeem S
Web: <https://www.icessu.com/>
Email: info@icessu.com

ICESSU Scopus®
5th International Conference on Engineering, Social Sciences and Humanities (ICESSU-2025)
9th - 10th April 2025
Manila, Philippines

Contact Number: +91 93429 24779
ICESSU-2025
09th -10th April 2025
Manila, Philippines.

Implementation Paper Acceptance

Bridging Language Barriers: The Role of AI in Real-Time Multilingual Translation for Video Conferencing

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Abstract—With the rise of global communication the need for effective multilingual translation in video conferencing has become crucial to break language barriers. Traditional video conferencing tools lack real time multilingual translation capabilities, limiting themselves in international collaborations. We present *Bridging language barriers: The role of AI in real-time multilingual translation for video conferencing a video conferencing system with live audio translation. This system facilitates communication between two people who speak different languages bridging the language barriers. Using AI based language models, our system captures live audio, process translations, and delivers them in real time reducing the latency up to 4-5 seconds and sometimes 3 second. The platform makes use of Neural Machine Translation NMT Algorithm which is a type of deep learning architecture which can learn to translate language.*

Index Terms—Video Conferencing, AI, WebRTC, Real-time Language Translation, MERN Stack, NLP, Facial Recognition, Noise Cancellation

I. INTRODUCTION

In today's globalized world, video conferencing has become an essential tool for communication, facilitating interactions across geographical and linguistic barriers. However, as the demand for international collaboration increases, existing video conferencing platforms face significant limitations, particularly in the areas of multilingual communication, background noise interference, and latency. Effective communication in multilingual settings is critical not only for businesses but also for educational institutions, healthcare providers, and diplomatic missions, where precision and clarity in communication are paramount. Unfortunately, current platforms offer limited support for real-time translation, often leading to misunderstandings and inefficient collaboration when participants speak different languages.

Moreover, the challenges of communication are exacerbated by background noise and high latency. Background

Noise whether from bustling office environments, home settings, or public places can severely disrupt the flow of conversation, making it difficult for participants to follow and understand each other.

This research aims to overcome these limitations by developing a comprehensive video conferencing system that integrates real-time multilingual translation, advanced noise cancellation, and significantly reduced latency. By leveraging cutting edge AI and machine learning techniques, the system delivers accurate translations across multiple languages, intelligently filters out background noise, and ensures seamless, low latency communication. This platform empowers users to engage in conversations without concerns about language differences, distractions, or delays, fostering more productive collaboration across borders.

The platform is built using the MERN stack ensuring scalability, robustness, and high performance. Real-time communication is facilitated through WebRTC technology, allowing peer to peer connections that ensure smooth video and audio transmissions with minimal delay. For multilingual communication, the system uses Natural Language Processing(NLP) and Speech Recognition APIs for real-time translation, transcription, and meeting summarization.

Additionally, the system utilizes several state-of-the-art tools to further enhance its capabilities. Next.js is employed for building server-side rendered applications, improving performance and SEO. TypeScript ensures strong typing, enhancing code reliability and maintainability. LiveKit serves as the infrastructure for real-time video and audio, and Pusher powers real-time updates and bidirectional communication via WebSockets. The platform also

utilizes TailwindCSS for responsive, utility first design and PlanetScale for a highly scalable, distributed database solution. To manage the platform's data with precision and type safety, Prisma ORM is used, interfacing with MySQL for efficient data handling.

The platform's real-time language translation is powered by the Google-translate-browser API for accurate multilingual conversion, while the Web Speech API handles transcription and text-to-speech synthesis. OneAI's Summarizer provides intelligent meeting summaries, aiding productivity by delivering concise insights from lengthy conversations.

II. LITERATURE SURVEY

Area of Study	Paper Name (Year)	Problem Identified	Solution
WebRTC	WebRTC: APIs and RTCWEB Protocols of the HTML5 Real-Time Web (2012) [1]	High latency and poor quality communication.	Integrate for WebRTC low-latency video/audio.
Noise Suppression	Deep Complex Networks for Real-Time Noise Suppression (2019) [2]	Background noise disrupts conversations.	Use deep learning models for effective noise cancellation.
Facial Recognition	DeepFace: Closing the Gap to Human-Level Performance in Face Verification (2014) [3]	Insecure authentication and low engagement.	Implement facial recognition for secure login and emotion detection.
Real-Time Translation	Google's Multilingual Neural Machine Translation System (2017) [4]	Language barriers hinder communication.	Employ NMT for instant multilingual translation.
Automated Summarization	Towards a Real-Time Meeting Assistant (2005) [5]	Inefficient manual note-taking.	Use NLP and speech-to-text for automatic meeting summaries.
Low-Latency Models	Low-Latency Machine Learning Models for Real-Time Communication (2022) [6]	Delays in audio/video affect user experience.	Implement low-latency ML models for smooth communication.
Speech Translation	Latency-Optimized Speech	Poor translation accuracy in noisy environments.	Combine noise reduction with real-time

	Translation with Noise Reduction (2019) [7]	Environments.	Real-time translation.
Adaptive Filtering	Adaptive Filtering for Real-Time Noise Reduction (2017) [8]	Inconsistent audio quality across users.	Use adaptive filtering for dynamic noise suppression.
Translation Challenges	Challenges in Real-Time Machine Translation (2020) [9]	Trade-offs between speed and accuracy.	Optimize models for efficient, accurate translations.
Latency Minimization	Latency Minimization Techniques in Multilingual Communication (2018) [10]	High latency in multilingual settings.	Use optimized algorithms to minimize latency.

TABLE I
LITERATURE SURVEY

III. EXISTING SYSTEM

Current video conferencing platforms have made significant advancements in virtual communication, providing high-quality audio and video. However, their ability to address critical issues such as real-time multilingual translation, effective noise cancellation, and reduced latency is limited. Several studies highlight these shortcomings and the gaps in current systems.

- Some platforms have begun integrating translation features, such as Skype Translator and Google Meet's live captions, but these are rudimentary and often lack the precision and fluency needed for professional communication. Research by Zhang et al. (2021) and Lee Kim (2022) discusses how current machine translation models struggle with real-time processing, contextual accuracy, and handling of domain-specific jargon or regional dialects.

- Many platforms offer basic noise suppression features. These platforms employ AI-driven noise cancellation algorithms, but research by Kumar Gupta (2017) indicates that these systems struggle with non-repetitive and complex background noises, such as construction sounds or multiple overlapping voices in shared spaces. While noise reduction has improved, its effectiveness decreases when dealing with more unpredictable noise sources.

- Latency remains one of the most challenging problems in video conferencing, particularly when real-time multilingual translation is involved. The

time required for speech recognition, translation, and synthesis increases delays, causing unnatural pauses in conversations. According to Wang et al. (2021), many video conferencing platforms are unable to sufficiently reduce latency during complex tasks like translation, particularly in environments with suboptimal network conditions.

IV. METHODOLOGY

A. *Proposed System*

To address the challenges in the existing systems in video conferencing, this paper proposes a comprehensive, integrated system that combines AI-driven technologies and optimized algorithms. The core of the proposed system is an advanced multilingual translation engine, powered by a hybrid model that combines Neural Machine Translation (NMT) and Statistical Machine Translation (SMT). This hybrid approach leverages the strengths of both models: NMT excels at understanding context and semantics through deep learning, while SMT provides robust performance in handling domain-specific terminology and less common language pairs.

The system captures speech in real-time, converting it to text using advanced speech recognition models that utilize state-of-the-art techniques, such as end to end deep learning architectures. This process ensures high accuracy in transcribing spoken language, even in environments with varying levels of background noise. Following transcription, the text is translated into the target language with minimal delay, thanks to a carefully designed pipeline that minimizes processing time at each stage. The translation module is trained using vast multilingual datasets, including diverse sources like academic papers, conversational transcripts, and technical documents, to improve contextual understanding and maintain accuracy across various topics and dialects.

To tackle the persistent issue of background noise, the system incorporates AI-driven noise cancellation algorithms that operate in tandem with the translation module. The noise cancellation feature employs advanced deep learning models such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) to effectively filter out unwanted noise from audio streams in real-time. These models are designed to recognize and separate human speech from various noise sources such as traffic, keyboard typing, and environmental sounds while preserving speech clarity. The system leverages a combination of

frequency domain analysis and time frequency masking techniques, allowing it to adapt dynamically to changes in the acoustic environment.

Moreover, the integration of these noise cancellation algorithms is facilitated by a feedback loop that continuously analyzes audio input, enabling the system to adjust its filtering strategies in real-time. This ensures that users experience crystal-clear audio even in challenging settings, such as crowded offices or busy home environments.

Additionally, the proposed system emphasizes user engagement and interactivity. It includes features like real-time feedback on audio quality and translation accuracy, allowing users to rate their experiences and provide input for continuous improvement. Advanced sentiment analysis algorithms are integrated to gauge user emotions during conversations, offering insights that can enhance communication dynamics and foster a more interactive meeting atmosphere.

Finally, to ensure that the system is accessible and secure, it employs robust user authentication methods, including facial recognition and voice verification, to protect sensitive conversations and maintain user privacy. This multifaceted approach not only addresses the pressing challenges of existing video conferencing systems but also aims to create a more inclusive, efficient, and engaging virtual communication environment for users across the globe.

Existing Systems have used Statistical Machine Translation (SMT) Algorithm which states - Statistical Machine Translation (SMT) is an earlier approach to translation that relies on statistical models to translate text. It works by dividing sentences into smaller phrases and analyzing bilingual corpora to identify patterns and probabilities of phrase translations. SMT models generate translations based on learned probabilities, often using alignment techniques to determine how words in the source language correspond to words in the target language. While SMT can be effective and relatively easier to implement, it tends to produce less fluent translations due to its fragmented approach and reliance on predefined linguistic rules. Moreover, SMT struggles with handling complex sentence structures and context, leading to potential inaccuracies in translations. This resulted in making the model inaccurate and prone to latency letting the applications struggle in translation of languages. SMT can be

slower and more resource-intensive than other models.

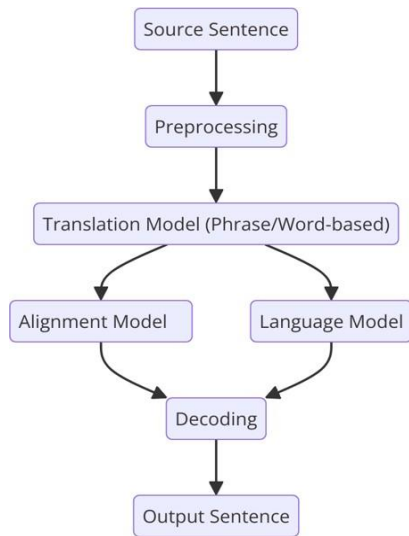


Fig. 1. STATISTICAL MACHINE TRANSLATION (SMT)

Alignment models given in above figure [Fig 1] are a sequence of increasingly complex models used in statistical machine translation to train a translation model and an alignment model, starting with lexical translation and moving to reordering and word duplication.

Language model is built from the output language monolingual data. The language model finds the candidate translations based on the translation language.

B. Algorithm to be used

As the SMT model discussed in existing systems struggles with handling complex sentence structures and context, leading to potential inaccuracies in translation and moreover is slower and more resource intensive. Neural Machine Translation (NMT) began to dominate.

1) *Neural Machine Translation (NMT)*: Neural Machine Translation (NMT) is a modern approach to translation that employs deep learning techniques, particularly neural networks, to process and translate text. NMT models analyze entire sentences as single units rather than breaking them down into smaller parts, allowing for better contextual understanding. This method often uses attention mechanisms, which help the model focus on relevant parts of the input sentence when generating translations. As a result, NMT tends to produce more fluent and coherent translations, even for complex sentence structures. NMT systems require substantial amounts of training

data but have demonstrated significant improvements in translation quality, making them suitable for real-time applications and diverse languages.

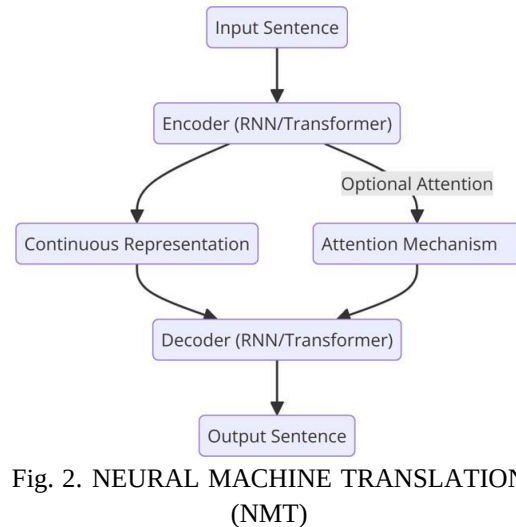


Fig. 2. NEURAL MACHINE TRANSLATION (NMT)

C. Application Areas

1) *Startups and MNCs with International Work Culture:*

- **Seamless Collaboration:** In a global business landscape, startups and multinational corporations (MNCs) frequently collaborate with teams and clients from various countries. A multilingual video conferencing system enables real-time communication, allowing employees to converse in their native languages without the need for extensive translation services. This fosters a more inclusive work environment and promotes effective collaboration.

- **Enhanced Productivity:** By reducing language barriers, these organizations can improve meeting efficiency and decision making processes. Employees can express their ideas more clearly and participate actively in discussions, leading to quicker resolutions and enhanced team synergy.

- **Global Talent Acquisition:** MNCs can broaden their recruitment efforts by hiring talent from diverse linguistic backgrounds. The ability to conduct interviews and onboarding processes in multiple languages can attract a wider range of candidates and enhance the company's global presence.

2) *Freelancers with Daily Foreign Communication:*

- **Diverse Clientele Management:** Freelancers often work with clients from different countries, necessitating clear communication. A multilingual video conferencing system allows freelancers to

negotiate contracts, discuss project details, and provide updates in their clients' preferred languages, building stronger relationships and trust.

- **Flexibility in Communication:** With real-time translation capabilities, freelancers can adapt to the communication styles of their international clients. This flexibility can lead to a better understanding of project requirements and expectations, ultimately resulting in higher client satisfaction and repeat business.

3) *Content Creators & Live Streamers to Reach a Global Audience:*

- **Engaging a Wider Audience:** Content creators and live streamers can significantly expand their reach by communicating with viewers in multiple languages. A multilingual video conferencing system enables them to host interactive sessions, Q&A forums, and tutorials in various languages, catering to a global audience.
- **Real-Time Interaction:** By leveraging real-time translation features, content creators can engage with their audience during live broadcasts, responding to comments and questions from viewers in their native languages. This creates a more immersive and inclusive experience, encouraging audience participation and loyalty.
- **Collaborative Content Creation:** Content creators can collaborate with peers from different countries, enabling them to produce diverse content that reflects various cultures and languages. This collaboration not only enhances creativity but also broadens their audience base.

4) *Educational Organizations to Teach Courses to Multi-lingual Students:*

- **Inclusive Learning Environment:** Educational institutions can leverage multilingual video conferencing systems to accommodate students from diverse linguistic backgrounds. This ensures that all students can participate in discussions and learn effectively, regardless of their language proficiency.
- **Global Classroom Initiatives:** With the ability to offer courses in multiple languages, educational organizations can establish global classrooms where students from different countries can learn together. This fosters cross-cultural understanding and prepares students for an increasingly interconnected world.
- **Support for Remote Learning:** The rise of remote learning necessitates effective communication tools that can bridge language gaps. A multilingual

video conferencing system can facilitate interactive lessons, group projects, and presentations, enhancing the overall learning experience for students and educators alike.

5) *Corporate Meeting Platform:*

- **Seamless Communication:** Companies often require secure, reliable, and efficient communication tools for internal and external meetings. The platform enables corporate teams to conduct virtual meetings with improved audio-visual quality and integrated security features. Provides AI-driven noise cancellation for a clear communication experience. Real-time translation for multinational teams. Provides Automated meeting summaries for better documentation.

V. FIGURES AND TABLE

Feature	Existing Systems	Proposed System	Statistical Improvement
Latency (Average Delay)	10-15 seconds	Reduced to 4-5 seconds (sometimes even 3 seconds)	Up to 67% reduction in latency
Browser Compatibility	Often reliant on specific browser APIs	Works across all browsers without reliance on specific APIs	Increased accessibility across platforms
Transcription Accuracy	Often inaccurate, leading to higher error rates in translations	Utilizes NMT and SMT for state-of-the-art transcriptions	Significant improvement in transcription accuracy

TABLE II
COMPARISON OF EXISTING AND PROPOSED SYSTEMS

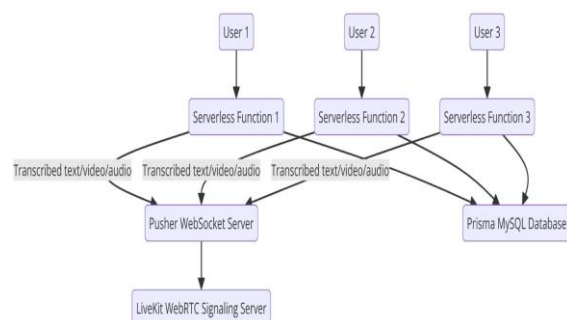


Fig. 3. ARCHITECTURE DIAGRAM OF SYSTEM

VI. CONCLUSION

The proposed multilingual video conferencing system significantly enhances real-time communication by addressing key challenges faced by existing platforms. By utilizing AI algorithms for improved transcription accuracy and reducing latency to 3-5 seconds, the system fosters clearer and more efficient conversations. Its compatibility across all browsers and innovative features, such as real-time meeting summarization and dynamic language switching, empower users with greater control and accessibility.

Furthermore, the integration of advanced noise cancellation techniques ensures that participants can communicate without distractions from background sounds, enhancing the overall quality of the interactions. This comprehensive solution not only bridges language gaps but also facilitates a more inclusive environment where users from diverse linguistic backgrounds can collaborate seamlessly.

In addition to the immediate advantages, the system is designed with scalability in mind, making it suitable for a wider range of applications from small startups to multinational corporations. The adaptive architecture allows for easy updates and the incorporation of new languages and dialects as they emerge, keeping pace with the ever-changing global landscape. The implementation of analytics features will enable organizations to gain insights into user engagement and communication patterns, which can inform strategies for improving collaboration and training. The continuous feedback loop from users will guide future enhancements, ensuring that the system remains relevant and effective in meeting user needs.

Ultimately, this solution sets a new standard for multilingual communication, paving the way for more effective cross-cultural collaborations in an increasingly interconnected world. As technology continues to evolve, so too will this system, driving innovation in virtual communication and making international interactions as straightforward and efficient as possible.

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Real-Time Multilingual Audio Translation and Transcription in Video Conferencing

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Abstract

This paper reports on a real-time multilingual speech-to-speech translation and transcription system deployed as part of a video conferencing application. Deepgram speech-to-text is used, and translation is handled using the Google Translate API. Pusher maintains smooth real-time updates, with LiveKit dealing with high-fidelity audio and video streaming in real time with little latency. The solution presented greatly mitigates language barriers, allowing smooth collaboration among speakers of different languages. This project showcases a feasible, scalable solution for enhancing communication in multilingual, diverse settings, opening the door to further innovations in AI-driven conferencing technology. It dynamically processes audio input, converts it into text, translates it into the desired language, and plays back the translated speech in real time. The architecture is highly scalable and optimized for minimal latency, ensuring a responsive and intuitive user experience. This project lays the foundation for future advancements in AI-driven video conferencing platforms, improving inclusivity and collaboration in multilingual scenarios.

Index Terms

Real-Time Translation, Multilingual Video Conferencing, Speech-to-Text (STT), Text-to-Speech (TTS), WebRTC Communication, Deepgram API, Google Translate API, LiveKit Integration, Pusher for Synchronization, Real-Time Subtitles, Low-Latency Communication

I. INTRODUCTION

With the recent world, this is the age of globalization, and the need for smooth communication is a must, even more so with real-time video conferencing where users from different linguistic backgrounds work together. With this requirement, our project combines the latest technologies to develop a multilingual, real-time audio translation and transcription system integrated in a video conferencing platform constructed with the MERN (MongoDB, Express.js, React, Node.js) stack. The system achieves high accuracy and efficiency by utilizing Deepgram for speech-to-text transcription and Google Translate API for real-time language translation. LiveKit manages low-latency video and audio streaming, maintaining smooth communication with minimal delay between various geographical locations. Pusher manages real-time event updates, making it possible for all participants to receive translated content and transcription at the same time. Supabase is also utilized for safe data storage and user authentication. The platform dynamically records, transcribes, translates, and plays back audio content in several languages, allowing participants to participate more effectively. Adaptive bitrate management and real-time event processing enable the system to operate at high performance even under fluctuating network conditions. The integration of features like real-time transcription, auto-detection of languages, and instant translation increases collaboration, and the system is suitable for remote teams, educational sites, and corporate communication. The scalable architecture of the project ensures that the system can manage multiple participants and various linguistic interactions effectively without compromising on quality. The blending of the technologies gives rise to a system that not only interprets speech but also retains contextual correctness, so that conversations are preserved in their true sense. Capable of transcribing, translating, and providing audio output in real-time, the platform enables users to carry out cross-language conversations with ease. Moreover, the system has been developed to support multiple participants at a time, thus being apt for mass application in virtual conferences, webinars, and international team meetings.

A. Motivation

The system supports effortless multilingual real-time communication during video conferencing through automatic spoken language translation into several target languages. It improves the user experience by eliminating the language barrier, thus making global communication more efficient and inclusive. It allows users to communicate in their preferred language. In addition to audio translation, chat translation and caption is also supported for an improved meeting experience. The objective involves combining Deepgram for speech-to-text transcription and Google Translate for translation of the transcribed text, ensuring that the spoken content is converted into text with precision and translated into the preferred language. LiveKit takes care of real-time audio and video streaming, whereas Pusher handles real-time event handling and synchronizing among the participants. This integration of technologies not only makes communication swift and effective but also inclusive so that people with different linguistic backgrounds can work together seamlessly.

II. LITERATURE SURVEY

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III. METHODOLOGY

A. System Architecture

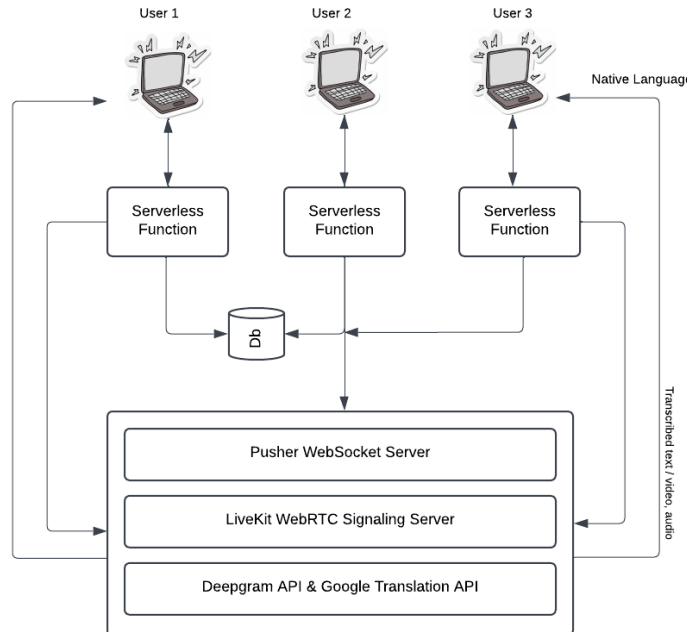


Fig. 1: Architecture Diagram of the System

1) **Client Side** : Provides an interactive interface for participants to communicate, communicate and receive translated messages in real time. Includes video and audio streaming, text displays for transcription, and translated content.

Technologies :

- React.js / Next.js for frontend.
- HTML, CSS, and JavaScript for UI design.

2) **LiveKit (Real-time Audio and Video Streaming)**: Livekit is a Live Streaming Platform that uses real-time WebRTC Technology to facilitate seamless audio and video communication among multiple users. It utilizes WebRTC (Web Real-Time Communication) to establish peer-to-peer connections and effectively transmit audio, video, and data streams.

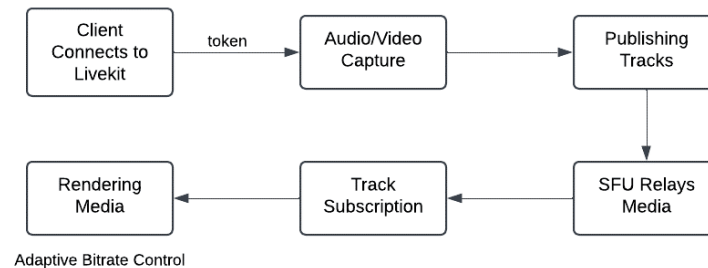


Fig. 2: LiveKit Workflow

LiveKit Workflow :

- 1) **Establishing connection** : A user opens the client application (browser or mobile app) and then the client requests a connection token from the backend to join a LiveKit room. The backend generates an access token with a unique identity using a LiveKit API key and secret. This token includes permissions such as **canPublish**, **canSubscribe**, and **canPublishData**. Then the token is sent back to the client. Client connects to the LiveKit server using WebRTC. ICE (Interactive Connectivity Establishment) protocol is used to negotiate and establish a connection between peers.
- 2) **Media Capture and Publishing** : Browser APIs capture audio and video using **getUserMedia()**. Media streams are encoded and prepared for transmission. Local audio and video tracks are created using LiveKit's **createLocalTracks()** API. Tracks are attached to a LocalParticipant object, representing the user. Media tracks are published to the LiveKit room using **room.localParticipant.publishTrack()**. LiveKit sends the audio and video streams to the SFU (Selective Forwarding Unit) server.
- 3) **Establishing WebRTC connection** :
ICE Negotiation - ICE protocol initiates the negotiation to establish a connection. STUN (Session Traversal Utilities for NAT) servers identify public IPs and assist in peer connection. TURN (Traversal Using Relays around NAT) servers relay media traffic if direct peer-to-peer communication fails.
SDP (Session Description Protocol) Exchange - Offer and answer messages containing media configuration (codecs, encryption, etc.) are exchanged. Once both peers agree on media configuration, the connection is established.
- 4) **Media Transmission and Routing** :
SFU (Selective Forwarding Unit) - LiveKit uses an SFU architecture where media streams from one participant are forwarded to others. SFU optimizes bandwidth by relaying only the required streams to each participant. Media is routed through LiveKit's SFU, ensuring scalability with multiple participants. Media streams are encoded using codecs such as VP8 (video) and Opus (audio).
Track Subscription - Other participants subscribe to the published audio and video tracks. Subscribed tracks are rendered on the respective participant's UI.
- 5) **Data Channel and Chat Support** : WebRTC's Data Channel is used for real-time exchange of text, messages, and other non-media data. Participants can send and receive text messages alongside audio and video.
- 6) **Bandwidth Management and Adaptive Bitrate Control** : LiveKit dynamically adjusts media quality to adapt to changing network conditions. If network bandwidth decreases, video quality is reduced to maintain smooth communication. Multiple versions of the same media stream are sent at different quality levels. SFU automatically switches between streams based on the receiving client's bandwidth.

3) **Deepgram (Speech to text processing)** : Deepgram is an AI-powered speech recognition API that converts audio into text in real time. It uses end-to-end deep learning models trained on vast datasets to ensure high accuracy, even in noisy environments or with different accents. Here's a detailed breakdown of how Deepgram processes and transcribes speech effectively:

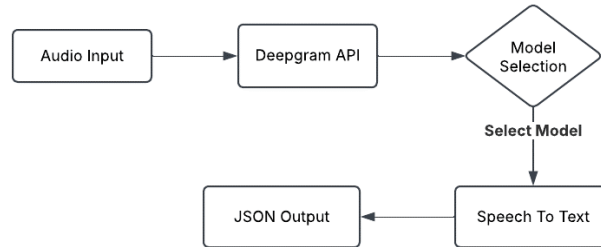


Fig. 3: Deepgram (Speech to text processing)

Deepgram Workflow :

- 1) **Audio Input and Streaming Setup** : The audio is captured in real time from the user's microphone during a video call. **LiveKit** handles the audio/video streams and transmits the audio in real time to Deepgram using WebRTC. The audio is transmitted over **WebSockets** to Deepgram's API endpoint ([wss://api.deepgram.com/v1/listen](https://api.deepgram.com/v1/listen)) for real-time transcription. WebSocket-based communication allows bi-directional, low-latency data flow.
- 2) **Audio Ingestion and Preprocessing** : Deepgram receives the audio stream in formats like PCM, WAV, or Opus. The system performs **signal preprocessing** to filter out background noise, normalize audio levels, and enhance speech clarity. Deepgram supports multiple audio channels, making it capable of identifying and processing individual speakers in a conversation.
- 3) **Speech-to-Text Conversion Using Deep Learning Models** : Deepgram uses convolutional neural networks (CNNs) and recurrent neural networks (RNNs) to analyze audio signals and extract relevant speech features. These models perform **phoneme-level analysis** and convert audio signals into linguistic representations. The system matches these patterns to produce high-accuracy transcriptions in real time. As audio data is streamed, Deepgram continuously updates the transcription with **partial results** that improve as the model processes more data. Deepgram's latency is as low as **300 milliseconds**, making it one of the fastest speech-to-text solutions available.
- 4) **Post-Processing and Output** : Deepgram applies additional post-processing techniques to improve transcription quality, including punctuation and capitalization. Final transcriptions are returned as structured JSON responses, which can be processed further for translation using the Google Translate API.

Deepgram's use of WebSockets ensures minimal overhead, enabling low-latency delivery of transcription results. WebSocket-based data streaming reduces the handshake delay common in HTTP requests. Deepgram's GPU-accelerated models process and transcribe audio in real time, allowing for seamless live transcription.

4) **Multilingual Translation (Google Translate API)** : The transcribed text is sent to the Google Translate API using an HTTP POST or GET request.

Request Format :

```

POST https://translation.googleapis.com/language/translate/v2
Body:
{
  "q": "Hello, how are you?",
  "target": "hi", // Target language code (e.g., 'hi' for Hindi)
  "source": "en", // Source language (op, auto-detect if not provided)
  "format": "text"
}
  
```

Response Format :

```
{
  "data": {
    "translations": [
      {
        "translatedText": "नमस्ते, आप कैसे हैं?",
        "detectedSourceLanguage": "en"
      }
    ]
  }
}
```

Google Translate API Workflow :

- 1) **Language Detection** - If no source language is specified, the API auto-detects the language. Language detection uses statistical models and pattern analysis.
- 2) **Translation Model** - Applies Neural Machine Translation (NMT) models. Context-aware translation improves accuracy by understanding surrounding text.
- 3) **Output Generation** - Returns translated text in the target language.

Text To Speech (TTS) Conversion : The translated text is sent to a TTS engine. Converts translated text into natural-sounding speech. Supports multiple languages and voice types. Response in audio format (MP3, WAV, etc.).

5) **Real time messaging and event handling (Pusher API) :** Pusher is used for real-time event-driven communication. It ensures that when a message is sent, it is instantly delivered to all connected clients. In this project, it is primarily used for: Real time chat updates, Synchronizing translated messages, Broadcasting messages to all participants in a chat.

User Sends Message - User types and sends the message in their native language.

Message Payload :

```
{
  "user": "Alice",
  "message": "Hello, how are you?",
  "language": "en",
  "targetLanguage": "hi"
}
```

The message is sent to the backend via an HTTP API or WebSocket.

Message is processed and translated - Backend receives the message. The original text is sent to Deepgram (if audio) or passed directly. Transcribed or original text is then sent to the Google Translate API. The translated message is generated in the target language.

Message sent to pusher - The translated message is sent to Pusher. Backend triggers a Pusher event:

```
pusher.trigger("chat-channel", "new-message", {
  user: "Alice",
  message: "नमस्ते, आप कैसे हैं?",
  originalMessage: "Hello, how are you?",
  language: "hi",
  originalLanguage: "en"
});
```

Channel - chat-channel

Pusher Broadcasts the Message - Pusher pushes the message to all connected clients. Clients subscribed to chat-channel receive the message instantly. Real-time update in the chat interface.

Client receives message and updates UI - All connected clients receive the translated message. WebSocket listener updates the UI with the translated text:

```
pusher.subscribe("chat-channel").bind("new-message", (data) => {
  displayMessage(data.user, data.message, data.language);
});
```

6) **tRPC (API communication)** : tRPC (TypeScript Remote Procedure Call) is used to enable seamless, type-safe communication between the frontend and backend without the need for REST API. It eliminates the need for writing multiple endpoints, handling API contracts, or manually defining request/response formats. The main goal is to ensure that the types defined on the backend automatically sync with the frontend, reducing the risk of errors and speeding up development.

7) **Security and access control:**

• **User Authentication and Session Management :**

Users log in using credentials (email, password) or OAuth (Google, etc.), verified against a secure database. After login, a session token (JWT) is issued and stored in cookies or local storage to maintain the user's session. Tokens have an expiry time, and refresh tokens ensure prolonged sessions without frequent re-login.

• **Role-Based Access Control (RBAC) for Meetings :**

- 1) **Participant:** Can join and interact with limited privileges.
- 2) **Guest:** May have view-only access.

IV. RESULTS

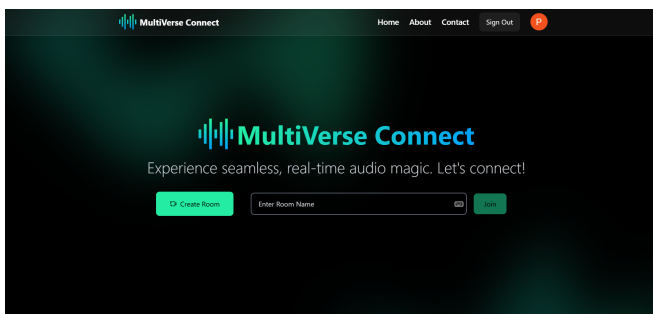


Fig. 4: Home Page

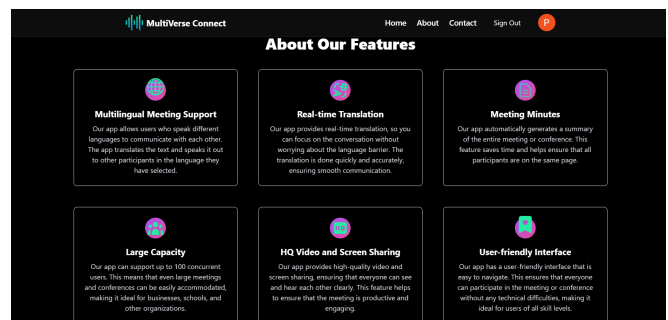


Fig. 5: About Us Page

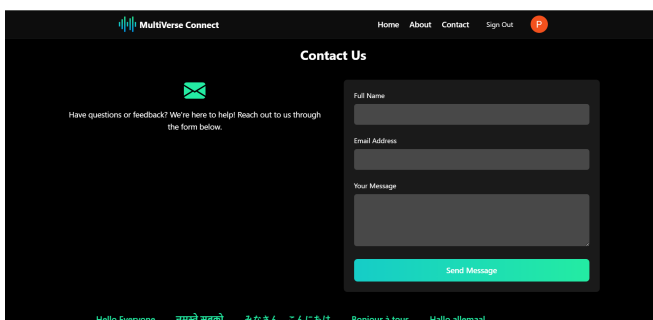


Fig. 6: Contact Page

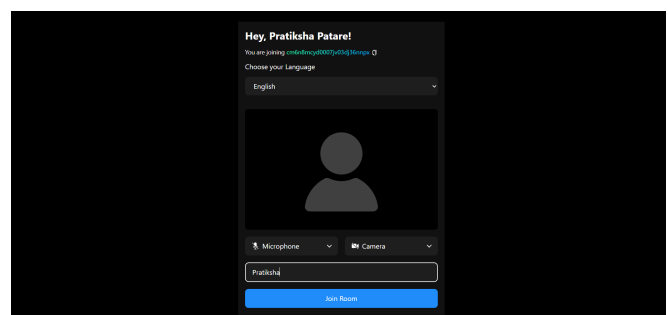


Fig. 7: Join Meeting

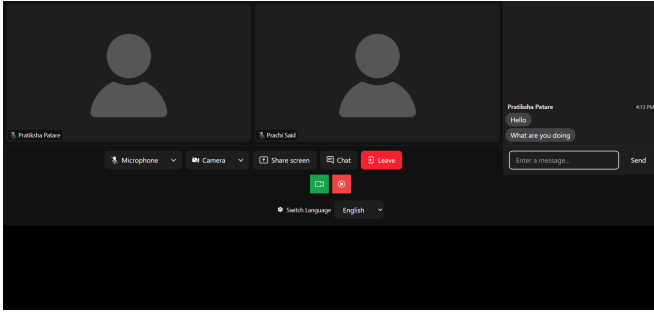


Fig. 8: English Chat from Sender End

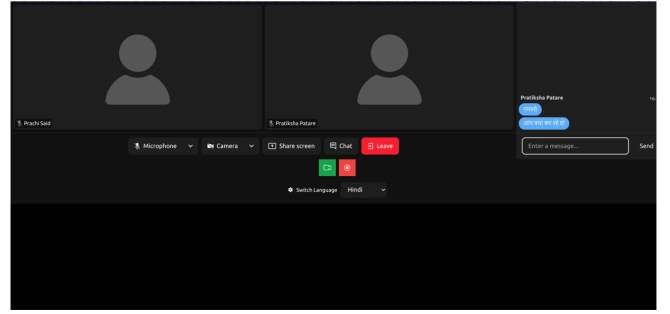


Fig. 9: Hindi Chat Received at Receiver End

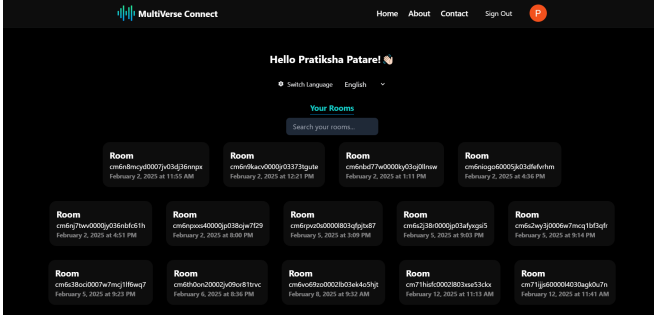


Fig. 10: Profile Details

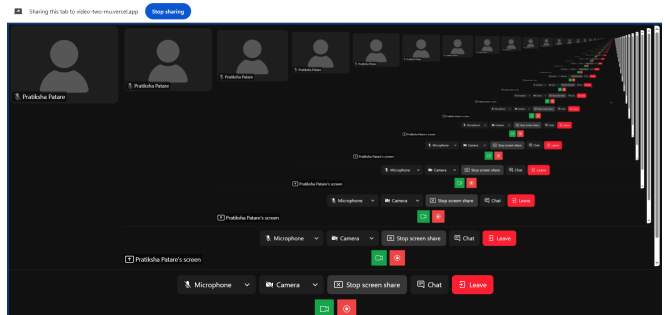


Fig. 11: Screen Sharing

V. PERFORMANCE COMPARISON CHART

Stage	Average Latency	Impact on System
Deepgram Speech-to-Text	100-200 (ms)	Minimal delay
Google Translate API	200-300 (ms)	Slight delay
Text-to-Speech Synthesis	150-250 (ms)	Moderate impact
Overall End-to-End Latency	2-3 (s)	Noticeable delay

TABLE I: Latency Comparison Chart

Condition	Video Quality (LiveKit)	Audio Quality (Deepgram)	Message Sync (Pusher)
High Bandwidth	HD Quality	High Fidelity	Instantaneous
Moderate Bandwidth	Medium Quality	Clear Audio	Slight Delay
Low Bandwidth	Low Quality	Degraded Audio	Noticeable Delay
Unstable Network	Frame Drops	Glitchy Audio	High Latency

TABLE II: Performance comparison based on bandwidth conditions

VI. CONCLUSION

The development of a real-time multilingual speech-to-speech translation and transcription system marks a significant advancement in overcoming language barriers in video conferencing. By integrating technologies such as Deepgram for speech recognition, Google Translate API for translation, and LiveKit for seamless audio-video streaming, this solution ensures low-latency and high-fidelity communication. The system's scalability and efficiency make it a viable option for multilingual collaboration, paving the way for future improvements in AI-driven communication platforms. With continued advancements, such technology can further enhance inclusivity and accessibility in global interactions.

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PRATIKSHA ASHOK PATARE



PRACHI SACHIN SAID



AFRIN RASHID SHAIKH



10.3 Appendix C

Plagiarism Report of project report.

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CHAPTER 11

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